Physics 1 – Lecture Notes

Dynamics I: Forces and Newton's Laws

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1 From kinematics to dynamics

Mechanics is traditionally divided into two parts:

- **Kinematics** $\rightsquigarrow$ the study of motion. Given the position of a body as a function of time, we compute its velocity and acceleration (or the other way round). We never asked what produces the acceleration.
- **Dynamics** $\rightsquigarrow$ the study of the *causes* of change in motion. Why does a stone fall with acceleration g?

The central concept of dynamics is the **force**. Kinematics gave us the language (position, velocity, acceleration); dynamics will tell us how forces determine accelerations, and therefore, through the kinematic equations, the whole motion.

2 Forces in nature

2.1 The four fundamental forces

There seem to be countless kinds of forces: the weight of a book, the friction of the floor, the tension in a rope, the push of a spring, the pull of a magnet, the lift on an aeroplane wing. Remarkably, physicists have found that all of them can be traced back to only **four fundamental forces** (or interactions):

Force	Acts on	Range	Where we meet it
Gravitational	all bodies with mass	infinite	weight, planets, tides
Electromagnetic	electric charges	infinite	almost everything in daily life
Strong	quarks, protons, neutrons	$\sim 10^{-15}$m	holds atomic nuclei together
Weak	all elementary particles	$\sim 10^{-18}$m	radioactive β decay, the Sun

The gravitational and electromagnetic forces have infinite range, so they act between macroscopic bodies and can be described by *classical* physics. The strong and weak forces act only at subnuclear distances and are intrinsically *quantum* phenomena; we will not meet them in this course.[1]

The four forces have enormously different strengths. Between two protons at nuclear distances, if we set the strong force equal to 1, the electromagnetic force is about 10^{-2}, the weak force about 10^{-6}, and gravity about 10^{-38}. Gravity is by far the weakest; we notice it only because it always attracts and adds up over the huge mass of the Earth, while electric charges come in two signs and mostly cancel.

2.2 Field forces and contact forces

From a practical point of view, it is useful to divide forces into two families:

1) **Field forces** $\rightsquigarrow$ there is *no need of contact*. The Earth pulls a falling apple without touching it; a magnet attracts a nail from a distance. The fundamental forces are field forces: each body creates a field around it, and the field acts on the other bodies.

2) **Contact forces** $\rightsquigarrow$ they *need contact*. Examples: someone pushing a box, a foot kicking a ball, a hand pulling or pushing a spring, the friction of the floor, the tension of a rope, and so on.

Intuition

Contact forces are not a fifth kind of force. When your hand pushes a box, the atoms of your hand never really "touch" the atoms of the box: the electrons of the two surfaces repel each other electromagnetically when they get very close. Contact forces are therefore *effective forces*: a convenient, macroscopic summary of an enormous number of microscopic electromagnetic interactions. In mechanics we do not need the microscopic picture; we only need to know how the contact force behaves on average.

2.3 Forces in classical mechanics

Intuitively, a force is a *push* or a *pull* on an object. And we know from experience that pushing or pulling an object changes its motion: a ball at rest starts rolling when kicked, a moving trolley stops when we hold it back, a ball on a string goes round in a circle instead of flying away. In short, a force causes a change of motion.

This intuition is a good starting point, but it is vague. What does "change of motion" mean exactly: a change of position, of velocity, of acceleration? How large is a force, and in which

[1] Strictly speaking, electromagnetism also has a quantum description, and gravity is described at the deepest level by Einstein's General Relativity. For the everyday phenomena of this course, Newton's gravity and classical electromagnetism are excellent approximations.

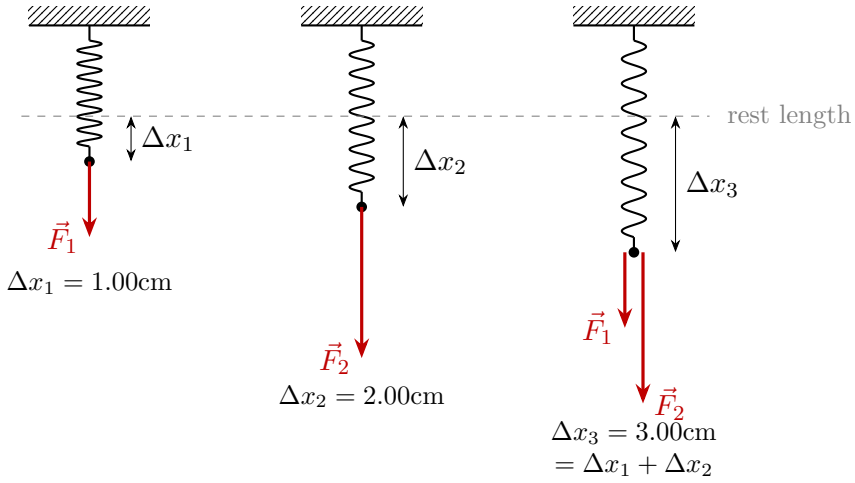
direction does it act? The definition of "force" in physics makes this intuition well defined. The rest of this lecture, and the next one, are devoted to building that definition step by step.

3 Force as a physical quantity

We want to establish "force" as a *physical quantity*: something that we can measure, with a number and a unit, and that obeys precise rules. The first question to settle is: *Is force a scalar or a vector?*

A practical way to measure a force is to apply it to a spring and look at how much the spring stretches. We take as an experimental fact that a spring stretches more when pulled harder, and that the elongation Δx is proportional to the size of the force, $|\vec{F}| \propto \Delta x$ (we will study this law, Hooke's law, in a later lecture). A spring calibrated in this way is a *dynamometer*, the simplest force-measuring instrument.

Now consider the following experiment, with a spring hanging from the ceiling.



(a) We pull the spring downwards with a force $\vec{F}_1$: it stretches by $\Delta x_1 = 1.00\text{cm}$.

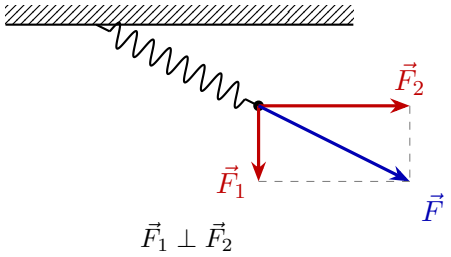
(b) We pull it downwards with a different force $\vec{F}_2$: it stretches by $\Delta x_2 = 2.00\text{cm}$. So $\vec{F}_2$ is twice as large as $\vec{F}_1$.

(c) We apply $\vec{F}_1$ and $\vec{F}_2$ *together*, both downwards: the spring stretches by

$$\Delta x_3 = 3.00\text{cm} = \Delta x_1 + \Delta x_2.$$

When the forces act in the same direction, the total force is the sum of the individual forces, exactly like adding numbers.

So far, forces could still be scalars. The decisive test comes when the two forces act in *different* directions.



(d) We apply $\vec{F}_1$ downwards and $\vec{F}_2$ horizontally, at the same time. The spring does not stretch vertically or horizontally: it tilts and aligns itself along a diagonal direction. Is its elongation Δx_4 equal to Δx_3? No! We measure

$$\Delta x_4 = 2.24\text{cm} \neq \Delta x_3 = 3.00\text{cm}.$$

Where does 2.24cm come from? If forces are vectors, the total force is the *vector sum* $\vec{F} = \vec{F}_1 + \vec{F}_2$. Since $\vec{F}_1$ and $\vec{F}_2$ are perpendicular, the magnitude of the sum follows from Pythagoras' theorem. Measuring forces in "centimetres of elongation", $|\vec{F}_1| \sim 1.00$ and $|\vec{F}_2| \sim 2.00$, so

$$|\vec{F}| = \sqrt{|\vec{F}_1|^2 + |\vec{F}_2|^2} \sim \sqrt{1.00^2 + 2.00^2} = \sqrt{5} \approx 2.24, \qquad |\vec{F}| \sim \Delta x_4.$$

This is exactly what we measure. Moreover, the spring points along the diagonal of the rectangle built on $\vec{F}_1$ and $\vec{F}_2$, i.e. along the direction of the vector sum.

Conclusion

The deformation of the spring depends on the *direction* of the force, and forces combine according to the rules of vector addition. Therefore **forces are vectors**: they have a magnitude, a direction, and they add as vectors. When several forces act on a body, their combined effect is that of a single force, the **net force** (or resultant),

$$\vec{F}_{\text{net}} = \sum_i \vec{F}_i = \vec{F}_1 + \vec{F}_2 + \dots$$

This is not obvious, and it is not a mathematical definition: it is an *experimental* fact about nature. Physics decides what kind of mathematical object best describes a quantity by looking at how that quantity behaves in experiments.

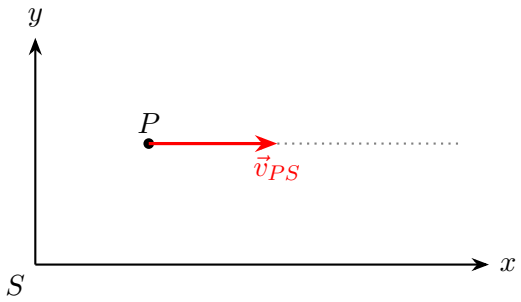
4 Newton's first law: the principle of inertia

The natural first question in dynamics is the simplest one: what happens to an object when *no force is impressed* on it?

For about two thousand years the accepted answer came from Aristotle: a body moves only as long as something pushes it; left alone, it will stop. This seems to agree with everyday experience: a book pushed across a table soon stops, and a cart stops when the horse stops pulling. It seemed that the origin of motion was someone acting of the object.

Galileo, in the seventeenth century, realised that this is an illusion. The book stops because of friction, which is a force as we ill see. If we make the table smoother and smoother, the book slides further and further. In the ideal limit of no friction at all, it would never stop, even if nobody is acting on it. This experimental fact contradicted Aristotle (and caused a lot of problems to the Galileo).

To see this more clearly, let us imagine a projectile moving in deep space, far from any star or planet. This is our best approximation of "no force impressed": there is no air, no friction, and gravity is negligible.



What do we observe? The projectile simply **continues to move**, in a straight line and at constant speed: it performs **uniform rectilinear motion**. We are in a reference frame S where P has a constant velocity $\vec{v}_0$. Here and in the following, $\vec{v}_{PS}$ denotes the velocity of P *relative to* the frame S (and similarly for positions). We can thus formulate the following:

Newton's first law (principle of inertia)

An object on which no force acts either remains at rest or continues to move in a straight line with constant velocity.

The tendency of a body to keep its state of motion is called **inertia**, hence the name of the law. Notice what the law says and what it does not say. A force is *not* needed to keep a body moving, supporting Galileo's observations and in contrast with Aristotele. On Earth, the best approximation of such situation is an ice-hockey puck on very smooth ice: after the hit, it glides almost in a straight line at almost constant speed, slowing down only because of the small friction of the ice.

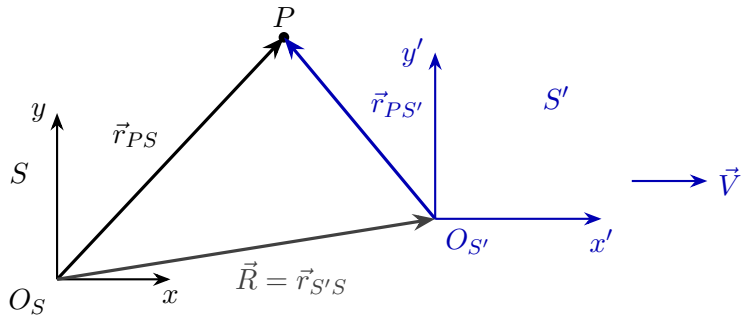
4.1 Rest frame

There is something puzzling in the law: "at rest" and "moving with constant velocity" are treated on the same footing. This is because, from the Galilean transformations, we can always find a reference frame in which the object is at rest.

In the frame S, the projectile moves with constant velocity, so its position and velocity are

$$\vec{r}_{PS}(t) = \vec{r}_{PS}(0) + \vec{v}_{PS}\, t, \qquad \vec{v}_{PS}(t) = \vec{v}_{PS}.$$

Consider now a second frame S', whose origin $O_{S'}$ has position $\vec{R} = \vec{r}_{S'S}(t)$ with respect to the origin O_S of S.



The Galilean transformation gives the position of P in S':

$$\vec{r}_{PS'}(t) = \vec{r}_{PS}(t) - \vec{R}(t).$$

Differentiating with respect to time,

$$\vec{v}_{PS'}(t) = \vec{v}_{PS}(t) - \vec{V}, \qquad \vec{V} = \frac{\mathrm{d}\vec{R}}{\mathrm{d}t} = \vec{v}_{S'S}(t),$$

where $\vec{V}$ is the relative velocity of $O_{S'}$ with respect to O_S.

The condition for P to be at rest in S' is that $\vec{v}_{PS'}(t) = \vec{0}$, which requires

$$\vec{v}_{S'S}(t) = \vec{v}_{PS}(t).$$

The observer who sees P at rest is simply an observer moving with the same velocity as P, $\vec{v}_{PS}(t) = \vec{v}_{PS}$. There is *no condition* on the position of the origin of S': Bob can be anywhere, as long as he moves exactly like P. Think of two cars driving side by side on a motorway at the same constant speed: each driver sees the other at rest.

Worked example

In the frame S a puck moves with $\vec{v}_{PS} = (3\,\hat{\imath} + 4\,\hat{\jmath})$m/s (speed 5m/s). An observer S' moving with $\vec{V} = (3\,\hat{\imath} + 4\,\hat{\jmath})$m/s sees $\vec{v}_{PS'} = \vec{v}_{PS} - \vec{V} = \vec{0}$: the puck is at rest. Another observer S'' moving with $\vec{V}'' = 3\,\hat{\imath}$m/s sees $\vec{v}_{PS''} = 4\,\hat{\jmath}$m/s: still uniform rectilinear motion, but in a different direction and with a different speed. Rest and uniform motion are the same physical state seen by different observers.

4.2 Inertial reference frames

We have just seen that the description of a force-free body depends on the observer. In some frames it is at rest, in others it moves uniformly. But there are also frames in which it does neither: seen from a merry-go-round or from a braking bus, a force-free body would appear to accelerate. The first law cannot be true for every observer. So it is really a statement about reference frames.

Inertial reference frame

The class of reference frames in which an object with no force impressed is at rest or moves with uniform rectilinear motion is called the class of **inertial reference frames**. Newton's first law states that **there exists** a reference frame in which a force-free object has zero acceleration, i.e. that inertial frames exist.

There are infinitely many inertial frames. If S is inertial and S' moves with *constant* velocity $\vec{V}$ with respect to S, then (as we saw in the last kinematics lecture) the two frames measure the same acceleration, $\vec{a}' = \vec{a} - \vec{A} = \vec{a}$ since $\vec{A} = 0$. A force-free body has $\vec{a} = 0$ in S, so it also has $\vec{a}' = 0$ in S': S' is inertial too.

Galilean transformations and Inertial frame

Galilean transformations with constant $\vec{V}$ transform an inertial frame into an inertial frame.

We can now restate the first law precisely. In an inertial frame:

(S1) No force $\implies$ the object is either at rest or moves with constant velocity.

We stress here that this statement is true only if we are in an inertial frame.

Notice also that this statement is an implication, and with implications we must be careful about which direction is true. There are two related statements.

The contrapositive is always true. An implication "$A \Rightarrow B$" is logically equivalent to "not $B \Rightarrow$ not A". Applied to (S1), in an inertial frame:

If our object has a velocity that is *not constant* in time $\implies$ there is a force acting upon it.

This is the most useful form of the first law in practice: *whenever we see a velocity change (in size or direction), we know a force is at work.* A car going round a bend at constant speed, a ball slowing down on the grass, the Moon orbiting the Earth: in all these cases the velocity changes, so a force must act.

The converse is *not* always true. The converse of (S1) would be "the object is at rest or moves with constant velocity $\Rightarrow$ no force acts on it". This is false. A book resting on a table is at rest, but two forces act on it: its weight, pulling it down, and the push of the table, holding it up. A car cruising at constant speed on a motorway is pushed forward by the road (thanks to the engine) and held back by air drag and friction. In both cases forces act, but they *balance*: their vector sum is zero.

Correct statement

In an inertial frame, constant velocity does not mean "no forces", but "**zero net force**":

$$\vec{v} = \text{const} \iff \vec{F}_{\text{net}} = \sum_i \vec{F}_i = \vec{0}.$$

Read with "no force" meaning "zero net force", the first law works in both directions. Whether one force or many forces act makes no difference to the motion; only their vector sum matters. This is why we needed to know that forces are vectors.

4.3 Non-inertial frames

Non-inertial frame

A **non-inertial frame** is a reference frame that accelerates with respect to an inertial frame.

In a non-inertial frame the first law fails: bodies can appear to accelerate without any force acting on them. A coffee cup on the smooth tray of an aeroplane slides backwards during take-off, although nobody pushes it; a passenger in a braking bus lurches forward. Seen from the ground (an inertial frame), nothing mysterious happens: the cup and the passenger simply tend to keep their velocity, while the aeroplane and the bus change theirs.

Is the ground really an inertial frame? Strictly speaking no, because the Earth rotates about its axis and orbits the Sun. But the corresponding accelerations are small (at most a few hundredths of a metre per second squared, compared with $g \approx 9.81\text{m/s}^2$), and for most laboratory experiments the Earth's surface can be treated as an inertial frame to an excellent approximation. We will come back to non-inertial frames, and to how to do physics in them, later in the course.

4.4 The first law in practice

Let us collect what the first law tells us, about the *absence* of forces, in a form we can use.

- If there is no external force, an object is either at rest or moves with constant velocity. The reference frames where this happens are known as **inertial reference frames**.
- If we know that there is *no force* impressed on our object, *but* we see the body accelerating, then we are in a **non-inertial** reference frame. The coffee cup on the tray is exactly this case.

How do we know whether our own frame is inertial? We can check it experimentally: we look at what happens to an *isolated* object, one that is far from everything else and on which no force acts. If it is at rest or moves with constant $\vec{v}$, we are in an inertial frame. After we have verified this once, we know for free that any other reference frame connected to ours via a Galilean transformation (with constant $\vec{V}$) is *also* an inertial frame.

From now on

From now on, unless stated otherwise, we will **always** work in an inertial reference frame. In such a frame:

$$\text{no force} \implies \text{no acceleration (at rest or } \vec{v} = \text{const}),$$
$$\text{acceleration} \implies \text{there is a force.}$$

5 Newton's second law

The first law tells us what happens when no force acts: nothing changes. It is a *qualitative* law. The second law is quantitative: it tells us exactly *how* the motion changes when a force acts.

5.1 Force and acceleration

We know that force is a vector quantity. Imagine pushing a puck on a horizontal, frictionless surface with a calibrated spring, and measuring its acceleration with the methods of kinematics. Experiments show that:

- we apply a force $\vec{F} \rightsquigarrow$ we observe an acceleration $\vec{a}$;
- we double the force, $2\vec{F} \rightsquigarrow$ we observe $2\vec{a}$: the acceleration doubles;
- if $\vec{F}$ changes direction $\rightsquigarrow \vec{a}$ changes direction too, and always points along $\vec{F}$.

In short, the acceleration is proportional to the force, and parallel to it:

$$\vec{F} \propto \vec{a}.$$

What if we apply two forces $\vec{F}_1$ and $\vec{F}_2$ at the same time? What really matters for the observed acceleration is their vector sum $\vec{F}_1 + \vec{F}_2$. In general, only the **sum of the applied forces** counts:

$$\sum_i \vec{F}_i \ \propto \ \vec{a}.$$

This is where the vector nature of force, which we established with the springs, becomes essential.

5.2 Inertial mass

The constant of proportionality between $\sum_i \vec{F}_i$ and $\vec{a}$ does not depend on the forces: it is a **property of the object**. We call it the **inertial mass** m_{in}.

Inertial mass

The inertial mass is an intrinsic property of an object, and it measures how much the object *resists* a change of its motion. The larger m_{in}, the smaller the acceleration $|\vec{a}|$ produced by the same total force $\sum_i \vec{F}_i$.

Everyday experience agrees: the same push that sends an empty shopping trolley rolling across the car park barely moves a fully loaded one. The loaded trolley has more inertia: a larger inertial mass. As a result we can formulate the second law of dynamics

Newton's second law

The total force applied to an object with inertial mass m_{in} is

$$\sum_i \vec{F}_i = m_{\text{in}}\,\vec{a}.$$

The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass.

Three remarks:

- **Many forces, one acceleration.** Many forces can be applied to an object, but there is **only one** acceleration $\vec{a}$, produced by their vector sum. It makes no sense to speak of "the acceleration due to force $\vec{F}_1$" and "the acceleration due to force $\vec{F}_2$" as two separate motions of the body.
- **Force is not the cause of motion.** It is the cause of *change of motion*. The second law relates force to acceleration, not to velocity. A body can move very fast with zero net force (a spaceship moving far away from any planet or star), and it can be momentarily at rest with a large net force (a ball at the top of its flight).
- **It is a vector equation.** It contains one equation for each component, $\sum_i F_{i,x} = ma_x$, $\sum_i F_{i,y} = ma_y$, which can be solved separately, exactly as we did in 2D kinematics.

From the second law, the dimensions of force are

$$\left[\vec{F}\right] = [m_{\text{in}}] \cdot [\vec{a}] = M \cdot \frac{L}{T^2} = \frac{M\,L}{T^2}.$$

The SI unit of force is the **newton** (N). One newton is the force that, applied to an object of inertial mass 1kg, produces an acceleration of 1m/s^2:

$$1\text{N} = 1\text{kg} \cdot \text{m/s}^2.$$

To get a feeling for it: the weight of a small apple (about 100g) is roughly 1N.

Worked example

A net horizontal force of 10N pushes a box of mass 2.0kg on a frictionless floor. Its acceleration is $a = F/m = 10/2.0 = 5.0\text{m/s}^2$. The same force on a 20kg box gives only

0.50m/s^2: ten times the mass, one tenth of the acceleration.

5.3 Mass calibration

How do we measure the inertial mass of an object? By comparing it with a **standard mass** m_s (by definition, the kilogram). Take the standard and a second object of unknown inertial mass m_{in}, and apply the *same* force $\vec{F}$ to both. From the second law,

$$\vec{F} = m_s\,\vec{a}_s \;\Rightarrow\; |\vec{F}| = m_s\,|\vec{a}_s| = m_s\,a_s, \qquad \vec{F} = m_{\text{in}}\,\vec{a}_{\text{in}} \;\Rightarrow\; |\vec{F}| = m_{\text{in}}\,|\vec{a}_{\text{in}}| = m_{\text{in}}\,a_{\text{in}}.$$

Equating the two expressions for $|\vec{F}|$:

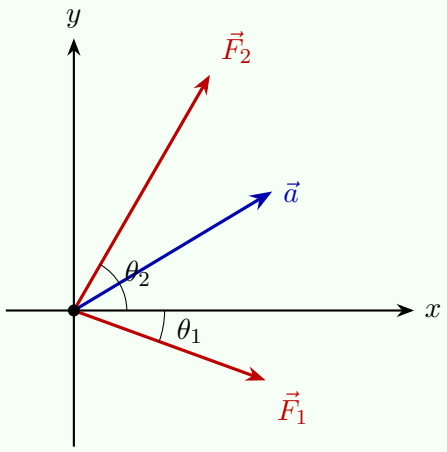
$$m_s\,a_s = m_{\text{in}}\,a_{\text{in}} \quad\Longrightarrow\quad \boxed{m_{\text{in}} = m_s\,\frac{a_s}{a_{\text{in}}}}$$

We only need to measure two accelerations, a purely kinematic task; we do not even need to know how large the force is. For example, if the standard kilogram accelerates at 3.0m/s^2 and the unknown object at 1.5m/s^2 under the same push, the object has $m_{\text{in}} = 1\text{kg}\times 3.0/1.5 = 2.0\text{kg}$.

5.4 Example: a puck pushed by two forces

Worked example (Example 5.1)

A hockey puck of mass $m = 0.30\text{kg}$ slides on a horizontal frictionless surface. Two forces act on it: $|\vec{F}_1| = 5.0\text{N}$ at $\theta_1 = -20°$ and $|\vec{F}_2| = 8.0\text{N}$ at $\theta_2 = 60°$ (angles measured from the x axis). Find the acceleration $\vec{a}$.



Strategy. Write the second law in components:

$$\sum_i \vec{F}_i = m\,\vec{a} \quad\Longrightarrow\quad \sum_i (F_{i,x}\,\hat{\imath} + F_{i,y}\,\hat{\jmath}) = m\,(a_x\,\hat{\imath} + a_y\,\hat{\jmath}) \quad\Longrightarrow\quad \begin{cases} \sum_i F_{i,x} = m\,a_x \\ \sum_i F_{i,y} = m\,a_y \end{cases}$$

Horizontal direction. With $F_{1,x} = |\vec{F}_1|\cos\theta_1$ and $F_{2,x} = |\vec{F}_2|\cos\theta_2$,

$$a_x = \frac{F_{1,x} + F_{2,x}}{m} = \frac{|\vec{F}_1|\cos\theta_1 + |\vec{F}_2|\cos\theta_2}{m}$$
$$= \frac{(5.0\text{N})\cos(-20°) + (8.0\text{N})\cos 60°}{0.30\text{kg}} \approx 29\text{m/s}^2.$$

Vertical direction. With $F_{1,y} = |\vec{F}_1| \sin\theta_1$ and $F_{2,y} = |\vec{F}_2| \sin\theta_2$,

$$a_y = \frac{F_{1,y} + F_{2,y}}{m} = \frac{|\vec{F}_1| \sin\theta_1 + |\vec{F}_2| \sin\theta_2}{m}$$
$$= \frac{(5.0\text{N}) \sin(-20°) + (8.0\text{N}) \sin 60°}{0.30\text{kg}} \approx 17\text{m/s}^2.$$

Note that $\sin(-20°) < 0$: $\vec{F}_1$ pulls slightly downwards, and the sign takes care of this automatically.

Magnitude and direction.

$$|\vec{a}| = \sqrt{a_x^2 + a_y^2} = \sqrt{29^2 + 17^2}\text{m/s}^2 \approx 34\text{m/s}^2,$$

$$\theta = \tan^{-1}\left(\frac{a_y}{a_x}\right) = \tan^{-1}\left(\frac{17}{29}\right) \approx 31°.$$

The acceleration points between the two forces, closer to the larger one, $\vec{F}_2$, exactly as the vector sum suggests.

6 Newton's third law

Forces never come alone. When you push a wall, the wall pushes back on your hands; when you jump, you push the ground down and the ground pushes you up. By observation: *every time a force is applied on an object by another object, the second object applies the same force on the first, but in the opposite direction.*

To keep the notation clear: object 1 applies a force $\vec{F}_{12}$ on object 2, i.e. $\vec{F}_{12}$ is the force applied *by* object 1 *on* object 2; similarly, $\vec{F}_{21}$ is the force applied on object 1 by object 2.



$\vec{F}_{21}$: on 1, by 2 $\qquad$ $\vec{F}_{12}$: on 2, by 1

Newton's third law

$$\vec{F}_{12} = -\vec{F}_{21}.$$

The two forces have the same magnitude and opposite directions. This is a **vector equality**. The two forces form an **interaction pair** (often called action and reaction).

Interaction pairs never cancel

The two forces of an interaction pair act on **different objects**. This is why they do not cancel each other: when we apply the second law to object 1, only $\vec{F}_{21}$ appears in the sum; $\vec{F}_{12}$ acts on object 2 and belongs to the second law for object 2. Forces can only cancel if they act on the same body.

Worked example

An apple of mass 0.20kg falls from a tree. The Earth pulls it down with a force of magnitude $mg \approx 1.96\text{N}$. By the third law, the apple pulls the Earth *up* with a force of exactly the same magnitude, 1.96N. Why does the Earth not move towards the apple? It does, but its acceleration is $1.96\text{N}/(5.97 \times 10^{24}\text{kg}) \approx 3 \times 10^{-25}\text{m/s}^2$.

6.1 The three laws together

The three laws are the fundamental basis to study how the motion of a body changes in time. Each one has its own role:

- **First law** $\rightsquigarrow$ it is not a quantitative law. It establishes the reference frames (the inertial ones) where the second and third laws can be applied.
- **Second law** $\rightsquigarrow$ it establishes the relation between force and acceleration, and allows us to define the inertial mass.
- **Third law** $\rightsquigarrow$ it establishes what happens when two bodies interact.

7 Mass vs. weight

In everyday language "mass" and "weight" are used as synonyms ("I weigh 70 kilos"). In physics they are different quantities, with different units.

Near the Earth's surface every object in free fall has the same acceleration, the gravitational acceleration

$$\vec{g} = -g\,\hat{\jmath}, \qquad g \approx 9.81\text{m/s}^2,$$

with the y axis pointing upwards. Apply the second law to a freely falling object: the only force acting on it is the gravitational force $\vec{F}_g$, and its acceleration is $\vec{a} = \vec{g}$. Therefore

$$\sum \vec{F} = m\,\vec{a} \quad \Longrightarrow \quad \vec{F}_g = m\,\vec{a} = m\,\vec{g}.$$

Gravitational force and weight

The **gravitational force** acting on an object of mass m is $\vec{F}_g = m\,\vec{g}$. The **weight** of the object is its magnitude:

$$|\vec{F}_g| = m\,|\vec{g}| = mg.$$

As a result we have the following:

- **Mass is a mass** $\rightsquigarrow$ it is measured in kilograms (kg).
- **Weight is a force** $\rightsquigarrow$ it is measured in newtons (N).

We know that the value of g can change: it is about 9.78m/s^2 at the equator and 9.83m/s^2 at the poles, it decreases with altitude, and on the Moon it is only 1.62m/s^2. Therefore the weight of an object also changes, depending on its geographical location.

- **Weight is not an inherent property** of the object: it depends on where the object is.

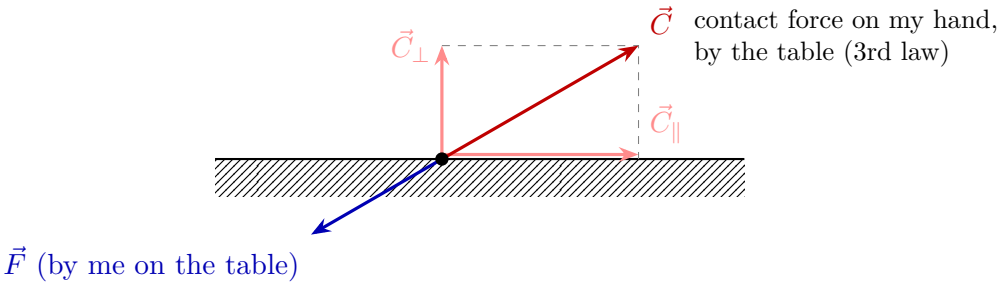
- **Mass is intrinsic.** Using the mass calibration procedure we always measure the same m_{in}, regardless of the geographical location (and even on the Moon or in deep space).

Worked example

An astronaut of mass 70kg has a weight of $70 \times 9.81 \approx 687$N on the Earth and $70 \times 1.62 \approx 113$N on the Moon, about one sixth. His mass is 70kg in both places: pushing him horizontally with the same force gives the same acceleration on the Earth and on the Moon. A bathroom scale actually measures a force and converts it into kilograms assuming $g = 9.81\text{m/s}^2$; on the Moon it would read about 11.6kg.

8 Contact forces

Contact forces occur when there is some contact between two objects. Let us study a simple situation: I push my hand obliquely against a table.

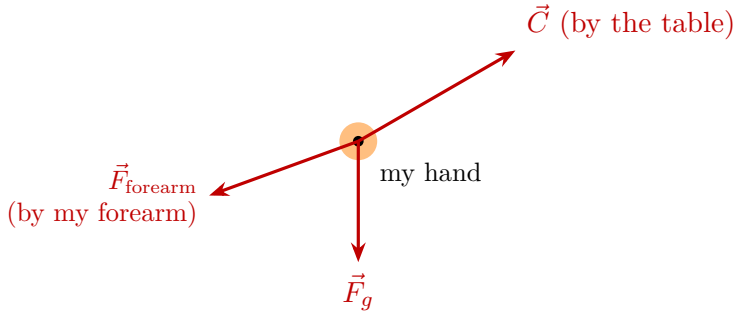


I apply a force $\vec{F}$ on the table. According to Newton's third law, the table applies a contact force $\vec{C} = -\vec{F}$ on my hand. It is convenient to split $\vec{C}$ into two components:

- $\vec{C}_\perp$, perpendicular to the surface $\rightsquigarrow$ the **normal force** $\vec{N}$. It prevents my hand from going *through* the table.
- $\vec{C}_\parallel$, parallel to the surface $\rightsquigarrow$ the **friction force** $\vec{f}$. It opposes the sliding of my hand along the table.

Normal force and friction are not two independent forces: they are the two components of the *same* contact force, decomposed along directions that are physically meaningful.

Now let us look at all the forces acting on my hand:



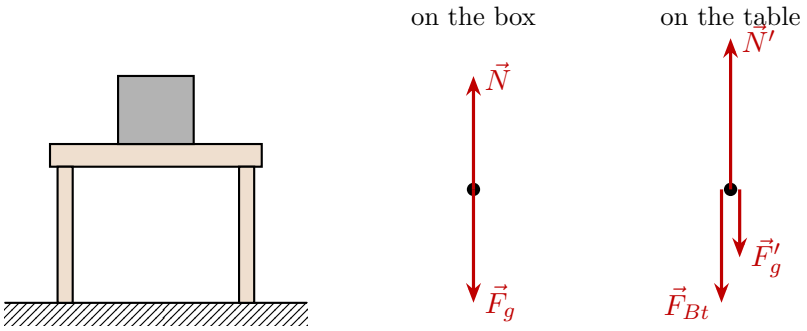
Three forces act on the hand: the contact force $\vec{C}$ caused by the table, its weight $\vec{F}_g$, and the force $\vec{F}_{\text{forearm}}$ applied by my forearm. If the hand is at rest, the second law requires $\vec{C}+\vec{F}_g+\vec{F}_{\text{forearm}} = 0$.

Contact forces adapt to the situation

Unlike the weight, which is always $m\vec{g}$, the contact force has no fixed formula. Its direction and magnitude depend on how hard and in which direction I push: they depend on the physical situation. Push harder, and the table pushes back harder; stop pushing, and the contact force disappears. We find contact forces by *applying Newton's laws*, not from a formula.

8.1 Free-body diagrams: a box on a table

Consider a box resting on a table, which in turn rests on the floor.



box on a table

The trick is to study the forces acting on the *two distinct objects* separately. For each object we draw a diagram where we *isolate* that object and draw all the forces acting on it, and only those.

Forces on the box. The weight of the box, $\vec{F}_g$, and the normal force $\vec{N}$ exerted by the table. The box is at rest, $\vec{a}_B = 0$, so

$$m\,\vec{a}_B = \sum_i \vec{F}_{i,B} = 0 \quad \Longrightarrow \quad \vec{N} + \vec{F}_g = 0 \quad \Longrightarrow \quad \vec{N} = -\vec{F}_g.$$

Forces on the table. The weight of the table, $\vec{F}_g'$; the normal force $\vec{N}'$ exerted by the floor (on its legs); and the force $\vec{F}_{Bt}$ exerted *by the box on the table*. The table is also at rest, $\vec{a}_T = 0$:

$$\sum_i \vec{F}_{i,T} = 0 \quad \Longrightarrow \quad \vec{N}' + \vec{F}_g' + \vec{F}_{Bt} = 0.$$

Connecting the two objects. The force of the box on the table and the force of the table on the box form an interaction pair. By the third law,

$$\vec{F}_{Bt} = -\vec{F}_{tB}, \qquad \text{and } \vec{F}_{tB} = \vec{N} \text{ (the force of the table on the box is the normal force).}$$

Therefore

$$\vec{N}' = -\vec{F}_g' - \vec{F}_{Bt} = -\vec{F}_g' + \vec{F}_{tB} = -\vec{F}_g' + \vec{N} = -\vec{F}_g' - \vec{F}_g.$$

The floor pushes the table up with a force equal in magnitude to the *total* weight of table and box: $|\vec{N}'| = (m_{\text{table}} + m_{\text{box}})\, g$. Every result makes physical sense: the table carries the box, and the floor carries both.

$\vec{N}$ and $\vec{F}_g$ are *not* an interaction pair

On the box, $\vec{N} = -\vec{F}_g$: equal and opposite. It is tempting to call this the third law, but it is not. The two forces act on the *same* object, and they are equal only because the box is not accelerating (second law). The third-law partner of $\vec{F}_g$ (the Earth pulling the box) is the box pulling the Earth; the partner of $\vec{N}$ (the table pushing the box) is $\vec{F}_{Bt}$ (the box pushing the table). If the table were in an accelerating lift, $\vec{N}$ and $\vec{F}_g$ would no longer be equal, but the third law would still hold.

Free-body diagram

This method is known as the **free-body diagram**:

1. isolate each object and draw *all* the forces acting *on* it (and none of the forces it exerts on others);
2. apply the second law to each object separately;
3. use the third law to connect the forces acting on the different objects.

We will use free-body diagrams in essentially every dynamics problem of this course. The next section shows how.

9 Applying Newton's laws

Solving a dynamics problem always follows the same recipe, built on the free-body diagram:

1. **Isolate the systems**: decide which object (or objects) you are studying.
2. **Find the external forces** applied on each system, and draw them in a free-body diagram.
3. Choose a coordinate system and write the second law component by component.
4. Solve the equations, and check that the result makes sense.

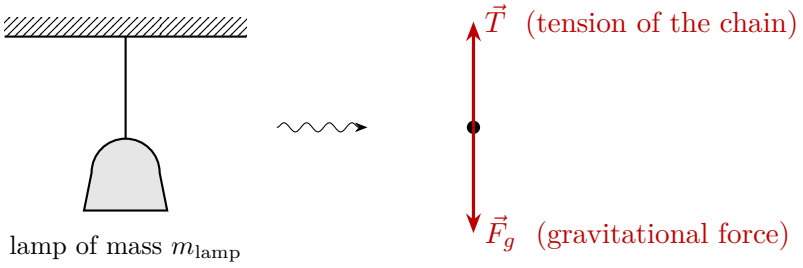
Two situations occur: the object is in equilibrium ($\vec{a} = 0$), or it is accelerating under a net force ($\vec{a} \neq 0$).

9.1 Particle in equilibrium

If an object is in **equilibrium**, i.e. at rest or moving with constant velocity, then

$$\vec{a} = 0 \quad \Longrightarrow \quad \sum_i \vec{F}_i = 0.$$

Consider a lamp of given mass m_{lamp} hanging from the ceiling by a chain.



Two forces act on the lamp: its weight $\vec{F}_g$ and the force exerted by the chain, the **tension** $\vec{T}$. Since the lamp is in equilibrium, $\vec{T} + \vec{F}_g = 0$, so $|\vec{T}| = m_{\text{lamp}}\, g$.

Tension of a rope. Ropes, strings, cables and chains appear in many problems. We will usually *neglect the mass of the rope*. Under this assumption:

- the **tension** is the magnitude of the force with which the rope pulls on the object attached to it;
- the **direction** of $\vec{T}$ is *away from* the object and always *parallel to the rope*. A rope can only pull, never push;
- a massless rope transmits the tension unchanged: the tension has the same magnitude at every point along it.

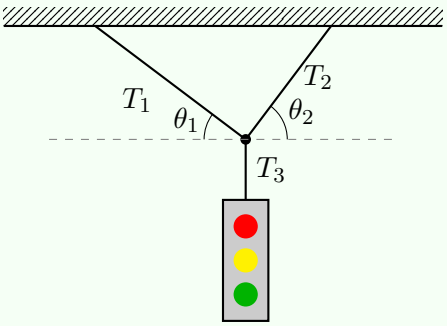
Tension and weight are not an interaction pair

$\vec{T}$ and $\vec{F}_g$ are equal and opposite, but they are **not** an action–reaction pair as described by the third law: $\vec{T}$ and $\vec{F}_g$ act on the *same object* (the lamp). They balance because the lamp is in equilibrium (second law). The third-law partner of $\vec{T}$ is the force with which the lamp pulls the chain down; the partner of $\vec{F}_g$ is the force with which the lamp pulls the Earth up.

9.2 Example: a traffic light

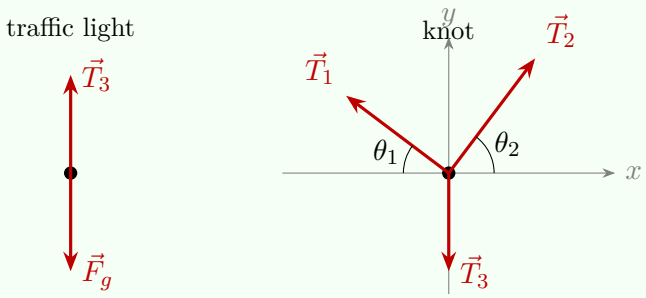
Worked example: a traffic light at rest

A traffic light of weight $W = 122\text{N}$ hangs from a cable 3, tied to two other cables 1 and 2 fastened to a support. Cables 1 and 2 make angles $\theta_1 = 37.0°$ and $\theta_2 = 53.0°$ with the horizontal. Find the tensions T_1, T_2, T_3 in the three cables.



Step 1: free-body diagrams. We isolate two systems: the traffic light, and the point

(knot) at which the three cables connect.



Notice that cable 3 pulls the traffic light *up*, and pulls the knot *down*: in both cases the tension points away from the object, along the cable.

Step 2: the traffic light. It is in equilibrium:

$$\vec{T}_3 + \vec{F}_g = 0 \quad \Longrightarrow \quad (T_3 - F_g)\,\hat{\jmath} = 0 \quad \Longrightarrow \quad T_3 = F_g = 122\text{N}.$$

Step 3: the knot. It is also in equilibrium. Measuring the direction of $\vec{T}_1$ from the positive x axis, its angle is $180° - \theta_1$.

Horizontal axis:

$$T_{2,x} + T_{1,x} = 0 \implies T_2 \cos\theta_2 + T_1 \cos(180° - \theta_1) = 0,$$

and since $\cos(180° - \theta_1) = -\cos\theta_1$,

$$T_2 \cos\theta_2 - T_1 \cos\theta_1 = 0.$$

Vertical axis:

$$T_{2,y} + T_{3,y} + T_{1,y} = 0 \implies T_2 \sin\theta_2 - T_3 + T_1 \underbrace{\sin(180° - \theta_1)}_{\sin\theta_1} = 0,$$

where, from the free-body diagram of the traffic light, $T_3 = F_g$.

Step 4: solve. We have two equations for two unknowns:

$$\begin{cases} (1) \quad T_2 \cos\theta_2 - T_1 \cos\theta_1 = 0 \\ (2) \quad T_2 \sin\theta_2 - F_g + T_1 \sin\theta_1 = 0 \end{cases}$$

From (1), $T_2 = T_1 \dfrac{\cos\theta_1}{\cos\theta_2}$. Substituting in (2):

$$\left(T_1 \frac{\cos\theta_1}{\cos\theta_2}\right) \sin\theta_2 + T_1 \sin\theta_1 = F_g \quad \Longrightarrow \quad T_1(\cos\theta_1 \tan\theta_2 + \sin\theta_1) = F_g,$$

so that

$$T_1 = \frac{F_g}{\cos\theta_1 \tan\theta_2 + \sin\theta_1} = \frac{122\text{N}}{\cos 37.0° \tan 53.0° + \sin 37.0°} = 73.4\text{N},$$

$$T_2 = T_1 \frac{\cos\theta_1}{\cos\theta_2} = 73.4\text{N} \frac{\cos 37.0°}{\cos 53.0°} = 97.4\text{N}.$$

Result: $T_1 = 73.4$N, $T_2 = 97.4$N, $T_3 = 122$N.

Check. Horizontally, $97.4 \cos 53.0° \approx 58.6$N and $73.4 \cos 37.0° \approx 58.6$N: they balance. Vertically, $97.4 \sin 53.0° + 73.4 \sin 37.0° \approx 77.8 + 44.2 = 122$N: the two upper cables together hold the full weight. The steeper cable (2) carries more of the load. Note also that $T_1 + T_2 > W$: tensions do *not* simply add up to the weight, because they are vectors.

9.3 Particle under a net force

If the net force is not zero, the object accelerates:

$$\sum_i \vec{F}_i \neq 0 \quad \rightsquigarrow \quad \sum_i \vec{F}_i = m\,\vec{a}.$$

We decompose Newton's law into each component:

$$\sum_i \left(F_{x,i}\,\hat{\imath} + F_{y,i}\,\hat{\jmath}\right) = m\,(a_x\,\hat{\imath} + a_y\,\hat{\jmath}) \quad \Longrightarrow \quad \begin{cases} \sum_i F_{x,i} = m\,a_x \\ \sum_i F_{y,i} = m\,a_y \end{cases}$$

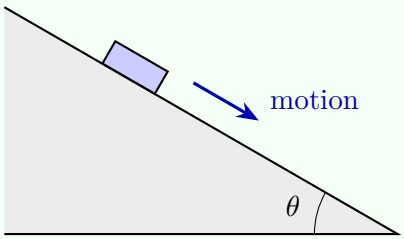
and solve each component separately.

9.4 Example: a car on a frictionless incline

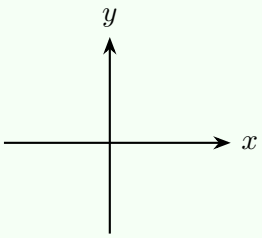
Worked example: sliding down an incline

A car of given mass m is left free on a frictionless inclined plane, which makes a given angle θ with the horizontal (think of a car on an icy road). Find its acceleration $\vec{a}$.

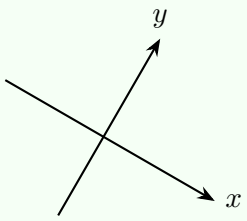
Choosing the coordinate system. Two choices are natural:



1st choice: horizontal / vertical

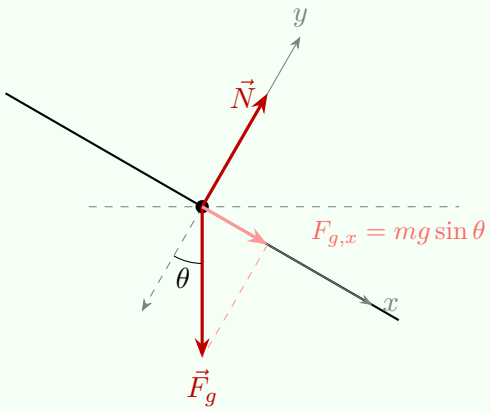


2nd choice: along / perpendicular to the incline



Both are possible choices of coordinate systems: the physics cannot depend on it. But one of them makes the calculation much easier. Since the car moves *along* the incline, the second choice (x along the incline, pointing down the slope, and y perpendicular to it) leaves only one non-zero component of the acceleration.

Free-body diagram. Two forces act on the car: its weight $\vec{F}_g = m\vec{g}$, vertical, and the normal force $\vec{N}$ of the incline, perpendicular to its surface. The angle between $\vec{F}_g$ and the $-y$ axis is θ, the same as the angle of the incline (the two angles have mutually perpendicular sides).



Second law, component by component.

- Along x: only the weight has an x component,

$$F_{g,x} = m\,a_x, \qquad F_{g,x} = |\vec{F}_g|\sin\theta = mg\sin\theta.$$

- Along y: the car does not leave the surface nor sink into it, so $a_y = 0$:

$$F_{g,y} + N_y = m\,a_y = 0.$$

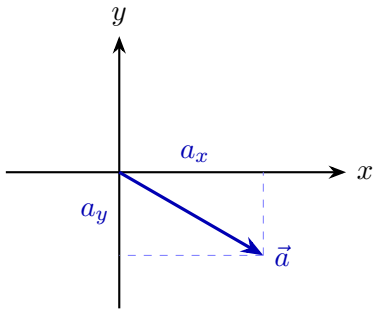
From the first equation, $mg\sin\theta = m\,a_x$, and the mass cancels:

$$a_x = g\sin\theta, \qquad a_y = 0, \qquad N = mg\cos\theta.$$

(The last result comes from the y equation, since $F_{g,y} = -mg\cos\theta$.)
Does it make sense? For $\theta = 0$ (flat road) $a = 0$: the car stays still. For $\theta = 90°$ (vertical wall) $a = g$: free fall. The acceleration does not depend on the mass, just as in free fall. Note also that $N = mg\cos\theta < mg$: on an incline the normal force is *smaller* than the weight. For $\theta = 30°$, $a = 9.81 \times 0.5 \approx 4.9\text{m/s}^2$.

The other coordinate system. In this coordinate system there is no vertical acceleration: the motion is purely along x. If we use instead the first coordinate system (horizontal and vertical axes), the same acceleration vector, of magnitude $g\sin\theta$ and pointing down the slope, has *two* non-zero components:

$$a_x = g\sin\theta\cos\theta, \qquad a_y = -g\sin^2\theta \qquad (\text{both } \neq 0!).$$



Coordinate system ≠ reference frame

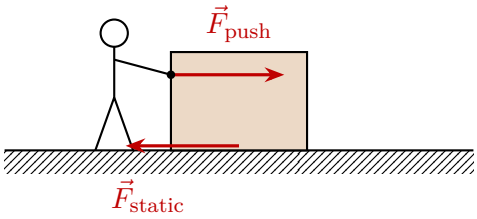
Changing from one set of axes to the other does *not* change the reference frame: both sets of axes are at rest in the same (inertial) frame, and every observer in that frame sees the car accelerate in exactly the same way. What changes is only how we *split* the same vector $\vec{a}$ into components. The physical result (magnitude $g \sin\theta$, direction down the slope) is identical; the second choice simply makes the algebra easier.

10 Friction

In the section on contact forces we saw that the contact force between two surfaces has a component parallel to the surface: the friction force. Friction is everywhere: it lets us walk, it makes cars brake, it stops a book sliding on a table. We now look at it more closely. There are two regimes, depending on whether the object is at rest or sliding.

10.1 Static friction

Push horizontally on a heavy box resting on the floor. If you do not push hard enough, the box does not move.



Since the box stays at rest, the net horizontal force is zero. Therefore a horizontal force must balance the push: the **force of static friction**, exerted by the floor on the box,

$$\vec{F}_{\text{push}} + \vec{F}_{\text{static}} = 0 \quad \Longrightarrow \quad \vec{F}_{\text{static}} = -\vec{F}_{\text{push}}.$$

If we increase $\vec{F}_{\text{push}}$ and the body is still not moving, then $\vec{F}_{\text{static}}$ must *also increase*, to counteract the pushing. Static friction is not an independent force with a fixed value: like the normal force, it *adapts* to the situation, taking exactly the value needed to prevent sliding.

It cannot grow without limit, though. There is a **maximum value** $F_{\max}$, which depends on the surface and on the object. If

$$|\vec{F}_{\text{push}}| > F_{\max},$$

static friction can no longer balance the push, and the body will start moving.

10.2 Kinetic friction

The friction that we experience *during* motion is known as **kinetic friction**, $\vec{F}_{\text{kin}}$. Now the box accelerates, and the second law reads

$$\vec{F}_{\text{push}} + \vec{F}_{\text{kin}} = m\,\vec{a}.$$

The kinetic friction force points opposite to the direction in which the box slides (here, opposite to the push).

10.3 How large is friction?

Experimentally, $\vec{F}_{\text{static}}$ and $\vec{F}_{\text{kin}}$ seem to depend on the weight of the object: a heavier box is harder to push. What is really relevant, however, is **not the weight but the normal force** $\vec{N}$ that the surface exerts on the object, i.e. how strongly the two surfaces are pressed together. On a horizontal floor the two have the same magnitude, but on an incline, or when someone pushes down on the box, they differ.

The magnitude of friction depends on $\vec{N}$ and on the nature of the two surfaces:

Laws of friction

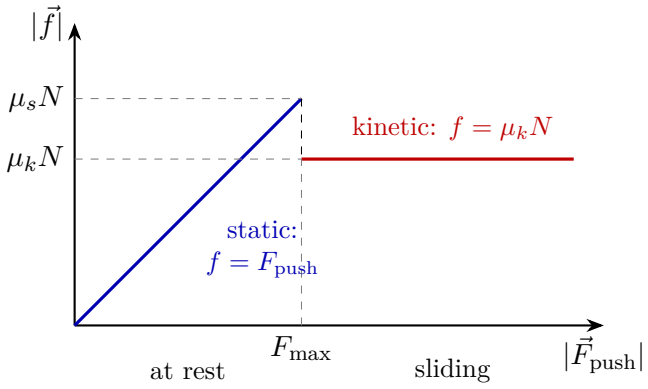
$$|\vec{F}_{\text{static}}| \leq \mu_s \, |\vec{N}| = F_{\max}, \qquad |\vec{F}_{\text{kin}}| = \mu_k \, |\vec{N}|,$$

where μ_s is the **coefficient of static friction** and μ_k the **coefficient of kinetic friction**. Both are pure numbers, which depend on the two surfaces in contact; usually $\mu_k < \mu_s$.

Inequality vs. equality

- Static friction is **not** always equal to $\mu_s|\vec{N}|$: that is only its *maximum*. Static friction "adapts" to the force we are applying, and takes any value between 0 and $\mu_s|\vec{N}|$. Writing $F_{\text{static}} = \mu_s N$ when the object is not on the verge of sliding is a very common mistake.
- Kinetic friction is (to a good approximation) **always** equal to $\mu_k|\vec{N}|$, independent of the speed and of the contact area.

The behaviour of friction as the push increases is summarised in the graph:



10.4 Direction of friction

The direction of friction is always opposite to the direction of the pushing/pulling force *in simple cases like the box above*. The precise rule is:

Direction of friction

Friction forces are always **parallel to the surface**, and **opposite to the motion** (or to the *potential* motion, in the case of static friction) **relative to the surface**.

"Relative to the surface" matters. When you walk, your foot pushes the ground backwards; the potential sliding of the foot is backwards, so static friction on your foot points *forwards*, and it

is this friction that moves you forward. A box on the floor of an accelerating lorry is carried forward by static friction, because it would otherwise slide backwards relative to the floor.

Worked example: pushing a box

A box of mass $m = 10\text{kg}$ rests on a horizontal floor with $\mu_s = 0.50$ and $\mu_k = 0.40$. On the flat floor $N = mg = 98.1\text{N}$, so

$$F_{\max} = \mu_s N \approx 49\text{N}, \qquad F_{\text{kin}} = \mu_k N \approx 39\text{N}.$$

- Push with 30N: since $30 < 49$, the box does not move, and static friction is 30N (*not* 49N!).
- Push with 60N: since $60 > 49$, the box starts sliding. Now friction is kinetic, and

$$a = \frac{F_{\text{push}} - \mu_k N}{m} = \frac{60 - 39.2}{10}\text{m/s}^2 \approx 2.1\text{m/s}^2.$$

Once the box moves, a push of only 39.2N is enough to keep it sliding at constant velocity: this is why it is harder to *start* moving a heavy piece of furniture than to keep it moving.

Worked example: measuring μ_s with an incline

Put a block on a plank and slowly raise one end. Along the incline, the weight pulls the block down with $mg\sin\theta$, and static friction holds it back; the normal force is $N = mg\cos\theta$. The block stays at rest as long as

$$mg\sin\theta \le \mu_s\, mg\cos\theta \quad \Longleftrightarrow \quad \tan\theta \le \mu_s.$$

It starts to slide at the critical angle θ_c with $\tan\theta_c = \mu_s$. Measuring θ_c gives μ_s directly, without knowing the mass. For $\mu_s = 0.50$, $\theta_c = \arctan 0.50 \approx 26.6^\circ$.

Surfaces	μ_s	μ_k
Rubber on dry concrete	1.0	0.8
Steel on steel	0.74	0.57
Wood on wood	0.25–0.5	0.2
Ice on ice	0.1	0.03
Teflon on Teflon	0.04	0.04

Typical (approximate) values of the friction coefficients.

11 Central forces: Newton's universal law of gravitation

Near the Earth's surface we have always written the gravitational force as

$$\vec{F}_g = m\vec{g} = -mg\,\hat{\jmath}.$$

This expression assumes that $\vec{g}$ is the same everywhere. But if we move far away from the Earth's surface (up a high mountain, to the altitude of a satellite, to the Moon) we will experience a *different* force: the pull of the Earth gets weaker with distance. The expression $\vec{F}_g = -mg\,\hat{\jmath}$ is just an approximation, valid close to the ground. Newton's great discovery was the general law,

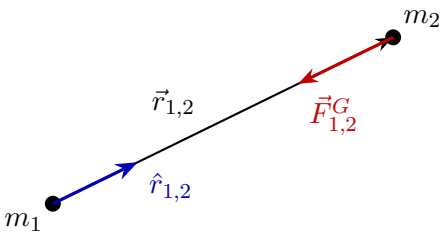
valid for any two bodies at any distance, which explains at the same time the fall of an apple and the orbit of the Moon.

11.1 Notation: position and unit vector

Consider two objects with masses m_1 and m_2. Let $\vec{r}_{1,2}$ be the vector that goes from object 1 to object 2. We write it as its length times a unit vector:

$$\vec{r}_{1,2} = |\vec{r}_{1,2}|\, \hat{r}_{1,2},$$

where $|\vec{r}_{1,2}|$ is the distance between the two objects, and the "hat" ($\hat{}$) denotes a **unit vector**: $\hat{r}_{1,2}$ has length 1 and points in the direction of $\vec{r}_{1,2}$, from 1 towards 2.



11.2 The law of universal gravitation

Newton's universal law of gravitation

The force on object 2 due to the gravitational interaction with object 1 is

$$\vec{F}^G_{1,2} = -\,G\,\frac{m_1\,m_2}{|\vec{r}_{1,2}|^2}\,\hat{r}_{1,2}, \qquad G = 6.67\times10^{-11}\,\frac{\mathrm{N\,m^2}}{\mathrm{kg^2}}.$$

Let us read the formula piece by piece:

- **The minus sign** means that the force on 2 points along $-\hat{r}_{1,2}$, i.e. from 2 back towards 1: gravity is always **attractive**.
- **The product** $m_1 m_2$: the force is proportional to both masses. Doubling either mass doubles the force.
- **The dependence** $1/r^2$: the force is inversely proportional to the *square* of the distance. At twice the distance the force is four times weaker; at ten times the distance, a hundred times weaker.
- **The direction** $\hat{r}_{1,2}$: the force acts along the line joining the two bodies. Forces with this property, which point along the line through a fixed centre and depend only on the distance from it, are called **central forces**.
- **The constant** G is the *gravitational constant*. It is very small ($\sim 10^{-11}$ in SI units), which is why gravity between everyday objects is imperceptible.

Worked example

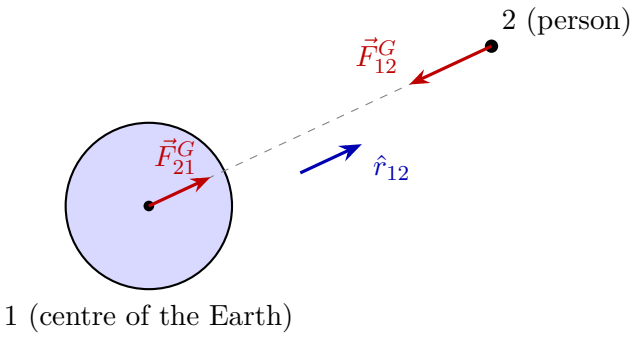
Two people of 70kg each, standing 1m apart, attract each other with a force

$$F = G\,\frac{m_1 m_2}{r^2} = 6.67\times 10^{-11}\,\frac{70\times 70}{1^2}\mathrm{N} \approx 3\times 10^{-7}\mathrm{N},$$

about the weight of a grain of sand. Gravity becomes important only when at least one of the two bodies is enormous, like a planet.

11.3 Example: the Earth and a person

Consider the gravitational force between the Earth (object 1) and a person (object 2). For a spherical body like the Earth, the distance is measured from its *centre*.



The force on the person due to the Earth, and the force on the Earth due to the person, are

$$\vec{F}^G_{12} = -\,G\,\frac{m_1 m_2}{r_{12}^2}\,\hat{r}_{12}, \qquad \vec{F}^G_{21} = -\,G\,\frac{m_2 m_1}{r_{12}^2}\,\hat{r}_{21}.$$

Since $\hat{r}_{21} = -\hat{r}_{12}$ (the unit vector from 2 to 1 is the opposite of the one from 1 to 2),

$$\vec{F}^G_{12} = -\vec{F}^G_{21}\,:$$

the law of gravitation satisfies **Newton's third law**. In particular

$$|\vec{F}^G_{12}| = |\vec{F}^G_{21}|\,:$$

the strength of the gravitational force on the Earth or on the person is the same!

So why does only the person fall? The person feels a strong attraction because of the *second* law. The same force produces very different accelerations on bodies of very different mass:

$$|\vec{F}^G_{12}| = m_2\,|\vec{a}_2| \quad\Longrightarrow\quad |\vec{a}_2| = G\,\frac{m_1}{r_{12}^2}, \qquad |\vec{F}^G_{21}| = m_1\,|\vec{a}_1| \quad\Longrightarrow\quad |\vec{a}_1| = G\,\frac{m_2}{r_{12}^2}.$$

The acceleration of the Earth contains the mass of the person, which is tiny compared with the mass of the Earth. So $|\vec{a}_1|$ is very small: the Earth experiences a negligible acceleration.

Worked example: where does g come from?

For a person on the Earth's surface, $m_1 = M_E = 5.97 \times 10^{24}$kg and $r_{12} = R_E = 6.37 \times 10^6$m. The acceleration of the person is

$$|\vec{a}_2| = G\frac{M_E}{R_E^2} = 6.67 \times 10^{-11}\,\frac{5.97 \times 10^{24}}{(6.37 \times 10^6)^2}\text{m/s}^2 \approx 9.8\text{m/s}^2.$$

This is exactly g! The constant acceleration of free fall is not a new law: it is the law of universal gravitation evaluated at the Earth's surface, $g = GM_E/R_E^2$. It also explains why g decreases with altitude (r grows). For a person of 70kg, the acceleration of the Earth towards the person is $|\vec{a}_1| = G \cdot 70/R_E^2 \approx 10^{-22}\text{m/s}^2$: completely unobservable.

12 Forces depending on velocity

All the forces we have met so far depend on the positions of the bodies, or adapt to the situation (normal force, static friction). There is another important family: forces that depend on the *velocity* of the object. Most of the time they appear due to the interaction with a **medium**, which can be a liquid or a gas. The most familiar example is **air resistance**: put your hand out of the window of a moving car, and you feel a force that grows as the car goes faster.

12.1 The resistive force

We call $\vec{R}$ the **resistive force** exerted by the medium. It depends on the velocity $\vec{v}$ of the object, and it is always opposite to the direction of the object's motion *relative to the medium*. In this respect it is analogous to kinetic friction; but it is different from friction, because its magnitude depends on the speed (kinetic friction, $\mu_k N$, does not).

For small objects moving slowly through a medium, the simplest model works well: the resistive force is proportional to the velocity,

Linear resistive force

$$\vec{R} = -\,b\,\vec{v},$$

where $\vec{v}$ is the velocity of the object relative to the medium, and $b > 0$ is a constant that depends on

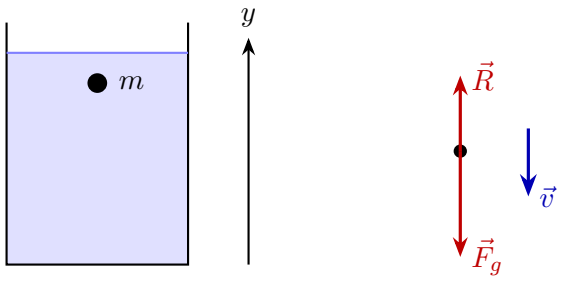
- the properties of the medium (water resists more than air, honey more than water);
- the shape and dimensions of the object.

The minus sign denotes that $\vec{R}$ is in the direction opposite to $\vec{v}$.

(For large objects at high speed, such as a skydiver or a car, the resistive force grows faster, roughly as v^2. The linear model is the simplest case, and it already shows all the essential physics.)

12.2 Example: a sphere falling in a liquid

A small sphere of mass m is released from rest at the surface of a liquid and sinks. We take the y axis pointing upwards.



Two forces act on the sphere: its weight $\vec{F}_g = -mg\,\hat{\jmath}$ and the resistive force $\vec{R} = -b\,\vec{v} = -b\,v_y\,\hat{\jmath}$. (We neglect the buoyancy of the liquid, which would simply reduce the effective weight.) Since the sphere is falling, $v_y < 0$, so $R_y = -b\,v_y > 0$: the resistive force points upwards, as in the free-body diagram.

Along y, the second law $\sum_i F_{i,y} = m\,a_y$ gives

$$-mg - b\,v_y = m\,a_y, \qquad a_y = \frac{\mathrm{d}v_y}{\mathrm{d}t},$$

that is

$$\frac{\mathrm{d}v_y}{\mathrm{d}t} = -g - \frac{b}{m}\,v_y.$$

This is an equation for the unknown function $v_y(t)$, which contains its own derivative (a *differential equation*). For a body starting from rest, $v_y(0) = 0$, its solution is

$$v_y(t) = \frac{mg}{b}\left(e^{-bt/m} - 1\right).$$

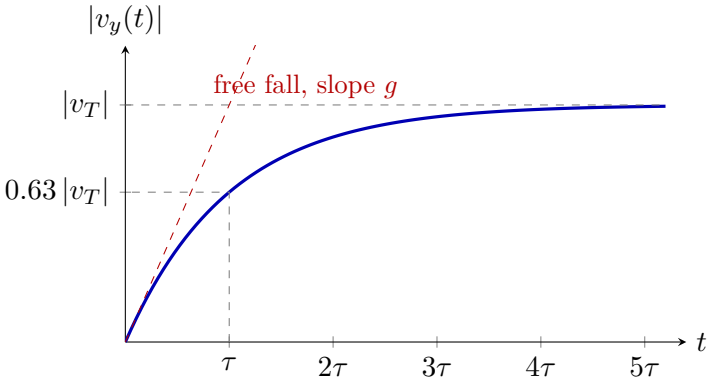
You do not need to know how to solve differential equations to check this: just substitute. The derivative of the proposed solution is $\mathrm{d}v_y/\mathrm{d}t = \frac{mg}{b}\cdot\left(-\frac{b}{m}\right)e^{-bt/m} = -g\,e^{-bt/m}$, while the right-hand side gives $-g - \frac{b}{m}\cdot\frac{mg}{b}\left(e^{-bt/m} - 1\right) = -g\,e^{-bt/m}$. They coincide. We also check the initial condition:

$$v_y(0) = \frac{mg}{b}(\underbrace{e^{-0}}_{1} - 1) = 0.\checkmark$$

What happens at long times? Since $\lim_{t\to+\infty} e^{-bt/m} = 0$,

$$v_y(\infty) = \frac{mg}{b}\,(0 - 1) = -\frac{mg}{b} \equiv v_T.$$

The velocity does not grow forever: it approaches a constant value v_T, the **terminal speed** (negative here because the sphere moves downwards).



The quantity $\tau = m/b$ has the dimension of a time and is called the **time constant**: after a time τ the sphere has reached $1 - e^{-1} \approx 63\%$ of its terminal speed, after 5τ more than 99%. At the very beginning, when v is still small, the resistive force is negligible and the sphere falls with acceleration g (red line).

12.3 Terminal velocity

When we have a resistive force, it is common to reach a certain terminal speed. This happens when the resistive force perfectly balances the other possible forces: then the net force is zero, the acceleration is zero, and the velocity stops changing.

This gives a quick way to find the terminal velocity without solving the differential equation. The condition for terminal velocity is $\vec{a} = 0$:

$$\vec{F}_g + \vec{R} = m\,\vec{a} = 0 \quad \Longrightarrow \quad -mg\,\hat{\jmath} - b\,v_T\,\hat{\jmath} = 0 \quad \Longrightarrow \quad v_T = -\frac{mg}{b}.$$

The same result, in one line. Heavier objects (larger m) and objects that the medium resists less (smaller b) reach a larger terminal speed.

Worked example

A small sphere of mass $m = 2.00\text{g}$ is released from rest in a vessel filled with oil, and reaches a terminal speed of 5.00cm/s. Then

$$b = \frac{mg}{|v_T|} = \frac{(2.00\times10^{-3})(9.80)}{5.00\times10^{-2}}\text{N s/m} \approx 0.392\text{N s/m}, \qquad \tau = \frac{m}{b} = \frac{|v_T|}{g} \approx 5.1\times10^{-3}\text{s}.$$

The sphere reaches 90% of its terminal speed when $e^{-t/\tau} = 0.1$, i.e. at $t = \tau \ln 10 \approx 2.30\,\tau \approx 11.7$ms: almost instantly. A raindrop or a skydiver also reaches a terminal speed (about 55m/s for a skydiver in free fall, before the parachute opens), which is why falling from a great height is not infinitely fast.

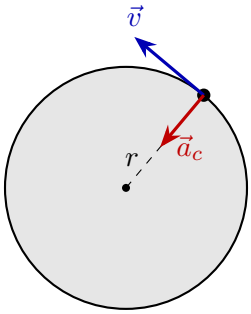
13 Dynamics of circular motion

13.1 Uniform circular motion and the force behind it

From kinematics we know that a body in uniform circular motion, with constant speed $v = |\vec{v}|$, has a **centripetal acceleration**

$$|\vec{a}_c| = \frac{v^2}{r},$$

whose direction is always towards the centre, perpendicular to the velocity, $\vec{a}_c \perp \vec{v}$.

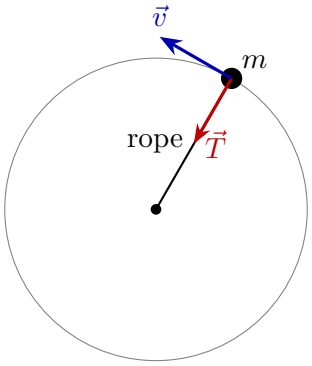


The centripetal acceleration is the only acceleration that the object is experiencing. The second law then tells us that

$$\sum_i \vec{F}_i = m\,\vec{a}_c :$$

if we have circular motion, there **must be a force** acting on our system, directed towards the centre, of magnitude mv^2/r. Without it, there is no centripetal acceleration and thus no circular motion.

Consider a ball of mass m tied to a rope and whirled in uniform circular motion on a horizontal frictionless table. The force responsible for the centripetal acceleration is the **tension** of the rope, which pulls the ball towards the centre.

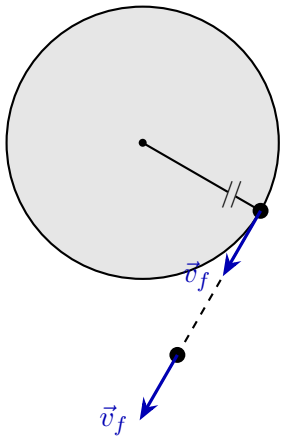


As we know, forces are the cause of the *change* in motion. Here the tension from the rope is the cause of the change in the *direction* of the velocity. It never changes the speed, because it is always perpendicular to the velocity.

"Centripetal force" is not a new kind of force

There is no special "centripetal force" to add to the free-body diagram. The word *centripetal* describes the *role* played by ordinary forces (a tension, the normal force, friction, gravity) whose sum points towards the centre. In a free-body diagram we draw only the real forces; then we require that their sum equal mv^2/r, towards the centre.

What if the rope breaks? If the rope breaks, there is no more tension, hence no more centripetal acceleration. The object no longer experiences a (horizontal) force, and by the first law it continues with **constant velocity** $\vec{v}_f$, the velocity it had at the instant the rope broke. It flies off *along the tangent* to the circle, not radially outwards.

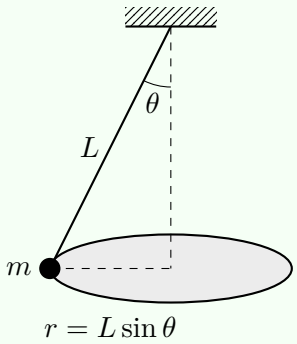


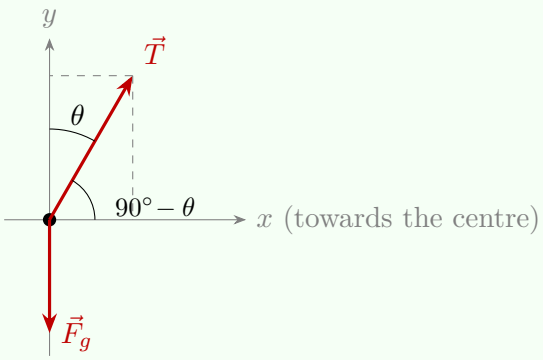
rope breaks ⇒ no tension
⇒ no force ⇒ constant $\vec{v}_f$

13.2 Example: the conical pendulum

Worked example: the conical pendulum

A small ball of mass m hangs from a string of length L. It moves with constant speed $v = |\vec{v}|$ in a horizontal circle of radius r, so that the string sweeps out the surface of a cone, making an angle θ with the vertical. Given L, m, r and θ, find $|\vec{v}|$.





Free-body diagram. Only two forces act on the ball: the tension $\vec{T}$ along the string and its weight $\vec{F}_g$. We take the x axis horizontal, pointing towards the centre of the circle, and y vertical. The tension makes an angle θ with the vertical, hence $90° - \theta$ with the horizontal.

Horizontal direction. The ball moves in a circle, so the net horizontal force must provide the centripetal acceleration:

$$\sum_i F_{i,x} = m\,a_x = m\,\frac{v^2}{r} \quad \Longrightarrow \quad |\vec{T}|\cos(90° - \theta) = m\,\frac{v^2}{r},$$

and since $\cos(90° - \theta) = \sin\theta$,

$$|\vec{T}|\sin\theta = m\,\frac{v^2}{r}.$$

Vertical direction. The ball does not move up or down, so

$$\sum_i F_{i,y} = m\,a_y = 0 \quad \Longrightarrow \quad |\vec{T}|\sin(90° - \theta) - mg = 0,$$

and since $\sin(90° - \theta) = \cos\theta$,

$$|\vec{T}|\cos\theta = mg \quad \Longrightarrow \quad |\vec{T}| = \frac{mg}{\cos\theta}.$$

Solve. Inserting $|\vec{T}|$ into the horizontal equation,

$$\frac{mg}{\cos\theta}\sin\theta = m\,\frac{v^2}{r} \quad \Longrightarrow \quad v^2 = r\,g\tan\theta \quad \Longrightarrow \quad |\vec{v}| = \sqrt{r\,g\tan\theta},$$

and with $r = L\sin\theta$,

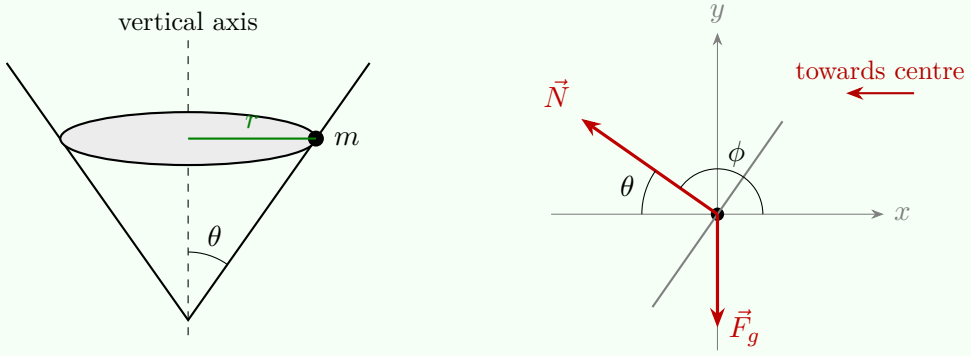
$$\boxed{|\vec{v}| = \sqrt{L\,g\,\sin\theta\,\tan\theta}}$$

The mass cancels: a heavy and a light ball on strings of the same length, swinging at the same angle, move at the same speed. For example, with $L = 1.0\text{m}$ and $\theta = 30°$: $r = 0.50\text{m}$, $v = \sqrt{1.0 \times 9.81 \times 0.5 \times 0.577} \approx 1.7\text{m/s}$, and one revolution takes $2\pi r/v \approx 1.9\text{s}$.

13.3 Example: an object sliding in a circle inside a cone

Worked example: circular orbit inside a cone

A small object of mass m slides without friction on the inside surface of a cone with vertical axis, whose wall makes an angle θ with the vertical. It performs uniform circular motion with speed $v_0 = |\vec{v}|$ in a horizontal circle. Find (1) the radius r of the circular path and (2) the time of one revolution.



Free-body diagram. Two forces act on the object: its weight $\vec{F}_g$ and the normal force $\vec{N}$, perpendicular to the wall of the cone (there is no friction). The object is on the right-hand side of the cone, so the centre of its circle is in the $-x$ direction.
Geometry. The wall makes an angle θ with the vertical, so its normal makes an angle θ with the horizontal. Measured from the positive x axis, the direction of $\vec{N}$ is therefore $\phi = 180° - \theta$.
Horizontal direction. The centripetal acceleration points towards the centre, i.e. along $-x$:

$$\sum_i F_{i,x} = m\,a_x = -m\,\frac{v_0^2}{r} \quad\Longrightarrow\quad |\vec{N}|\cos\phi = -m\,\frac{v_0^2}{r}.$$

Since $\cos\phi = \cos(180° - \theta) = -\cos\theta$, multiplying by (-1):

$$(1) \qquad |\vec{N}|\cos\theta = m\,\frac{v_0^2}{r}.$$

This equation contains two unknown variables, $|\vec{N}|$ and r: we need a second equation.
Vertical direction. There is no vertical motion:

$$\sum_i F_{i,y} = 0 \quad\Longrightarrow\quad |\vec{N}|\sin\phi - mg = 0,$$

and since $\sin\phi = \sin(180° - \theta) = \sin\theta$,

$$|\vec{N}|\sin\theta = mg \quad\Longrightarrow\quad |\vec{N}| = \frac{mg}{\sin\theta}.$$

1) Radius. Inserting $|\vec{N}|$ in equation (1):

$$\frac{mg}{\sin\theta}\cos\theta = m\,\frac{v_0^2}{r} \quad\Longrightarrow\quad r\,\frac{1}{\tan\theta} = \frac{v_0^2}{g} \quad\Longrightarrow\quad \boxed{r = \frac{v_0^2}{g}\tan\theta}$$

2) Time of revolution. In one revolution the object covers $2\pi r$ at constant speed v_0:

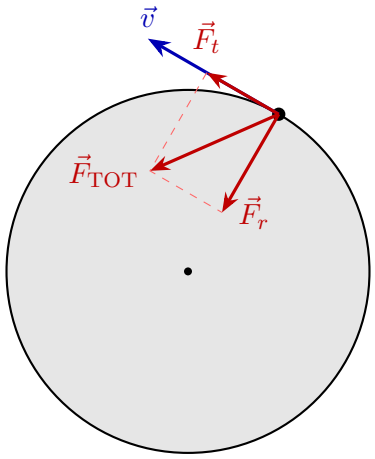
$$T = \frac{2\pi r}{v_0} = \frac{2\pi v_0 \tan\theta}{g}.$$

Does it make sense? A faster object circles higher up the cone, where the radius is larger ($r \propto v_0^2$). The mass cancels again. For $v_0 = 2.0\text{m/s}$ and $\theta = 30°$: $r = (4.0/9.81) \times 0.577 \approx 0.24\text{m}$ and $T \approx 0.74\text{s}$; the normal force is $N = mg/\sin 30° = 2\,mg$, twice the weight.

Notice that the two examples have the same structure as a car on a banked curve: in both cases a force that is not horizontal (the tension, the normal force) has a horizontal component that provides the centripetal force, and a vertical component that balances the weight.

13.4 Non-uniform circular motion

So far the speed on the circle was constant, and the net force pointed exactly towards the centre. If there is a change *also in the speed*, then, by the second law, there must be a force component in the **tangential** direction as well. In general the total force $\sum_i \vec{F}_i$ on a body moving on a circle has two components:

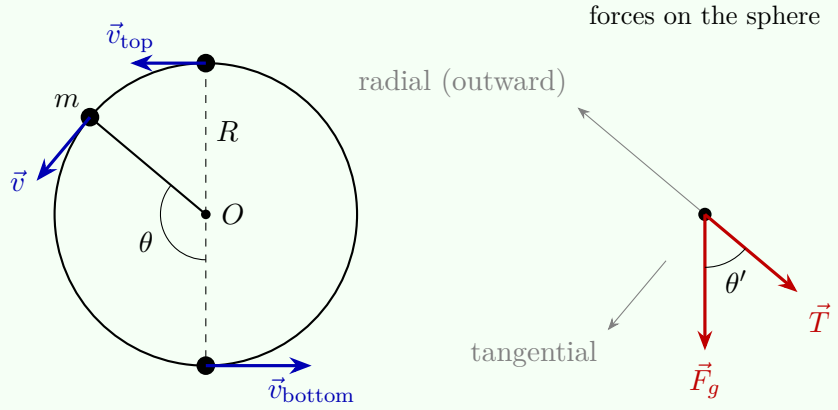


- a **radial** component $\vec{F}_r$, towards the centre, which changes the direction of the velocity: $\sum F_r = m\,v^2/r$;
- a **tangential** component $\vec{F}_t$, along the velocity, which changes the speed: $\sum F_t = m\,a_t$, with $a_t = \mathrm{d}|\vec{v}|/\mathrm{d}t$.

This is the dynamical counterpart of what we found in kinematics: the acceleration has a centripetal part $a_c = v^2/r$ and a tangential part a_t.

Worked example: a ball on a cord in a vertical circle

A small sphere of mass m is attached to a cord of length R and whirled in a *vertical* circle about a fixed point O. Its speed is *not* constant: it is larger at the bottom, v_{bottom}, and smaller at the top, v_{top}. How does the tension of the cord change along the circle?



Setting up. We measure the position of the sphere with the angle θ between the cord and the *downward* vertical ($\theta = 0$ at the bottom, $\theta = 180°$ at the top). Two forces act on the sphere: the tension $\vec{T}$, along the cord towards O, and the weight $\vec{F}_g$. We use two axes adapted to the circle: the **radial** axis, along the cord, and the **tangential** axis, along the velocity. The tension is purely radial; the weight makes an angle $\theta' = 180° - \theta$ with the cord direction towards O, so it has both a radial and a tangential component.

Along the tangential axis. Only gravity contributes:

$$mg\cos(\theta - 90°) = m\,a_t, \qquad \cos(\theta - 90°) = \sin\theta \quad \Longrightarrow \quad a_t = g\sin\theta.$$

The tangential acceleration depends on the position. It vanishes at the top and at the bottom ($\sin\theta = 0$), and it is largest when the cord is horizontal. It always points "downhill" along the circle: gravity slows the sphere down on the way up and speeds it up on the way down.

Along the radial axis. Taking the direction towards the centre as positive, the net radial force must equal $ma_c = mv^2/R$:

$$T + mg\,\underbrace{\sin(\theta - 90°)}_{-\cos\theta} = m\,a_c = m\,\frac{v^2}{R} \quad \Longrightarrow \quad T - mg\cos\theta = m\,\frac{v^2}{R},$$

so that

$$T = mg\left(\frac{v^2}{gR} + \cos\theta\right).$$

The tension is **not constant**: it depends on v and on θ. At the top and at the bottom of the circle:

$$T_{\text{top}} = mg\Big(\frac{v_{\text{top}}^2}{gR} + \underbrace{\cos 180°}_{-1}\Big), \qquad T_{\text{bottom}} = mg\Big(\frac{v_{\text{bottom}}^2}{gR} + \underbrace{\cos 0°}_{1}\Big).$$

The tension is **larger at the bottom**, for two reasons: there gravity points opposite to $\vec{T}$ (away from the centre), so the cord must also balance the weight; and the speed is larger.

A consequence. A cord can only pull, so $T \geq 0$ everywhere. At the top this requires $v_{\text{top}}^2/(gR) \geq 1$, i.e. $v_{\text{top}} \geq \sqrt{gR}$. If the sphere is slower than this at the top, the cord goes slack and the sphere leaves the circle. For example, with $m = 0.20\text{kg}$, $R = 0.50\text{m}$, $v_{\text{top}} = 3.0\text{m/s}$ and $v_{\text{bottom}} = 5.3\text{m/s}$, one finds $T_{\text{top}} \approx 1.6\text{N}$ and $T_{\text{bottom}} \approx 13\text{N}$; the minimum speed at the top is $\sqrt{gR} \approx 2.2\text{m/s}$.

14 Dynamics in non-inertial reference frames

14.1 Newton's laws hold only in inertial frames

Let us recall what we know. In inertial frames Newton's laws hold, so if we observe an acceleration, $\exists\, \vec{a} \neq 0$, there must be a force. To understand whether we are in an inertial reference frame we need to do one thing:

- take an object on which no force is applied;
- check whether it is at rest or moving with constant $\vec{v}$.

If this is true, we are in an inertial reference frame. From such a frame, by Galilean transformations, we obtain infinitely many other inertial reference frames. In all of them, if we observe an acceleration, there is a force acting on our object.

In a **non-inertial** reference frame this statement is false! We can observe an acceleration *without* a force acting on the object, and

$$\sum_i \vec{F}_i = m\,\vec{a}$$

is not valid any more. Yet we often have to describe motion from accelerating frames: a train that speeds up, a car on a curve, a rotating platform, the Earth itself. What can we do?

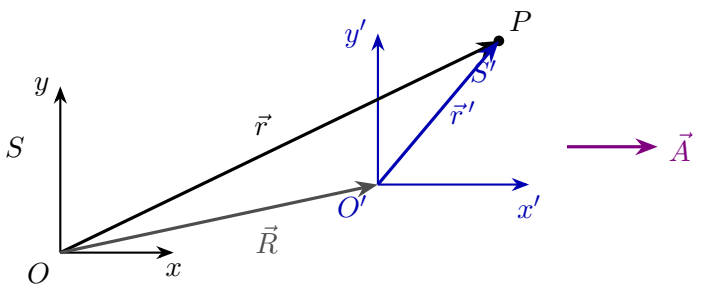
14.2 A linearly accelerating frame: the train

Put a ball on the (frictionless) floor of a train.

- **Train at constant speed.** The ball at rest stays at rest. Inside the train you cannot tell whether you are moving or not.
- **Train accelerating forward.** The ball starts rolling *backward*, although nobody pushes it: there is no horizontal force on it at all.

Is Newton wrong? No! The accelerating train is simply not an inertial frame. Seen from the platform (an inertial frame) the ball does nothing: it stays where it was, because no horizontal force acts on it. It is the train that accelerates forward underneath it.

Let S be the inertial frame (the platform) and S' the frame of the train, which translates (without rotating) with respect to S. As in the kinematics of relative motion, call $\vec{R}$ the position of the origin of S' seen from S, with velocity $\vec{V}$ and acceleration $\vec{A}$. For a point P:



$$\vec{r} = \vec{R} + \vec{r}' \quad \Longrightarrow \quad \vec{v} = \vec{V} + \vec{v}' \quad \Longrightarrow \quad \vec{a} = \vec{A} + \vec{a}', \qquad \vec{A} = \frac{\mathrm{d}\vec{V}}{\mathrm{d}t},$$

where primed quantities are measured in S'. (In the Galilean transformations of the kinematics lectures we had $\mathrm{d}\vec{V}/\mathrm{d}t = 0$; now we allow it to be non-zero.)

Case 1: train at constant speed, $\vec{A} = 0$. Then $\vec{a}\,' = \vec{a}$, so $m\vec{a}\,' = m\vec{a}$, and

$$\sum_i \vec{F}_i = \sum_i \vec{F}'_i,$$

where the left-hand side are the forces as seen from S and the right-hand side the forces as seen from S'. Both observers write the same second law. Any frame moving at constant velocity with respect to an inertial frame is itself inertial: no mechanical experiment done inside the train can reveal its uniform motion.

Case 2: train accelerating, $\vec{A} \neq 0$. Multiplying the relation between accelerations by m,

$$m\,\vec{a} = m\vec{A} + m\,\vec{a}\,'.$$

Since S is an inertial frame, the second law holds there, and we can write $m\vec{a} = \sum_i \vec{F}_i$. For the non-inertial frame we therefore get

Second law in a linearly accelerating frame

$$m\,\vec{a}\,' = \sum_i \vec{F}_i \underbrace{- m\vec{A}}_{\vec{F}_{\text{fict}}}.$$

In the accelerating frame, the body moves *as if* an extra force $\vec{F}_{\text{fict}} = -m\vec{A}$ acted on it. It is the "compensation" for the fact that the frame itself is accelerating. This is a **fictitious force**: it is not a real force, but something that we see emerge due to the choice of the reference frame. In a non-inertial reference frame, $m\vec{a}\,'$ corresponds to the physical forces *plus* an extra fictitious force.

For the ball on the floor of the train, no horizontal physical force acts, $\sum_i \vec{F}_i = 0$ horizontally, so

$$\vec{a}\,' = -\vec{A}\,:$$

it rolls backward, exactly as observed.

Properties of fictitious forces

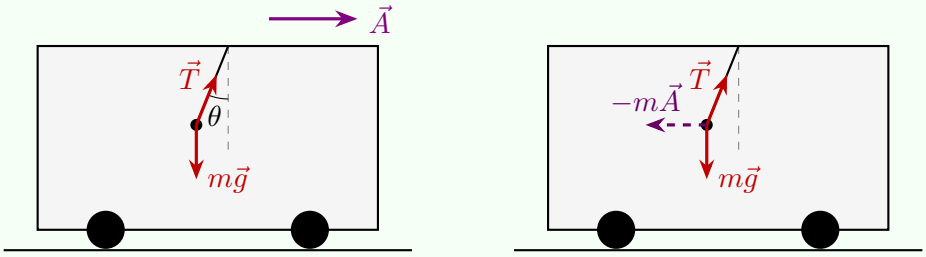
1. They are **not physical forces**: no body exerts them, they do not come from an interaction.
2. Hence the **third law is not satisfied**: there is no action–reaction pair.
3. They appear **only in non-inertial frames**; in an inertial frame they simply do not exist and must never be drawn.
4. They point **opposite to the acceleration of the frame**.
5. They are **proportional to the mass** of the body (like gravity).

"Fictitious" does not mean imaginary

The passenger in a braking bus *is* thrown forward, and a scale in an accelerating lift *does* read a different weight. What is fictitious is the *interpretation* as a force produced by something. Whether we use fictitious forces or not is a choice of frame: the two descriptions give the same physics. The golden rule: *either* work in an inertial frame, with real forces only and $\vec{a} \neq 0$; *or* work in the accelerating frame, with real + fictitious forces and the acceleration $\vec{a}\,'$ measured there. Never mix the two (for example, drawing $-m\vec{A}$ *and* writing $\sum \vec{F} = m\vec{A}$ counts the acceleration twice).

Worked example: a pendulum as an accelerometer

A small mass m hangs from the ceiling of a train that accelerates forward with constant $\vec{A}$. The string settles at an angle θ from the vertical. Find θ.



(a) seen from the platform (inertial)

(b) seen from the train (non-inertial)

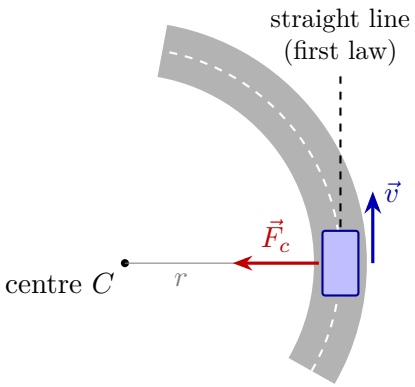
(a) Inertial frame. The mass accelerates with the train, $\vec{a} = \vec{A}$. Only two physical forces act, $\vec{T}$ and $m\vec{g}$, and their sum must give $m\vec{A}$ (horizontal):

$$T\sin\theta = mA, \qquad T\cos\theta - mg = 0.$$

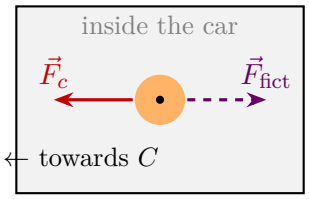
(b) Train frame. The mass is at rest, $\vec{a}\,' = 0$, so the forces including the fictitious one must balance: $\vec{T} + m\vec{g} - m\vec{A} = 0$, i.e. the *same* two equations.
Dividing: $\tan\theta = A/g$. For $A = 2.0\text{m/s}^2$, $\theta = \arctan(2.0/9.81) \approx 11.5°$. Measuring θ inside the train measures the train's acceleration.

14.3 The centrifugal force

A change in the *magnitude* of the train's velocity produced a fictitious force. A change in the *direction* of the velocity produces another one. Consider a passenger in a car that takes a curve of radius r at constant speed v: the passenger is moving inside the car in a circle, in uniform circular motion.



(a) seen from the ground (inertial): the seat/door pulls the passenger towards C: $|\vec{F}_c| = mv^2/r$



(b) seen from the car (non-inertial): the passenger is at rest, so $\vec{F}_c + \vec{F}_{\text{fict}} = 0$

Inertial frame (ground). By the first law the passenger tends to continue in a straight line. The car instead follows the circle, and the passenger "goes with it" only if a real force pulls her towards the centre: the friction force of the seat or, if that is not enough, the push of the door of the car. This real force plays the role of centripetal force:

$$|\vec{F}_c| = m\,\frac{v^2}{r} \qquad \text{(directed towards the centre).}$$

Nothing pushes the passenger outwards: it is the car that "turns into" her.

Reference frame of the car, S'. The passenger is at rest, yet we know that friction is acting on her, in order to have circular motion. The car frame is non-inertial: with respect to the ground it has the centripetal acceleration $\vec{A}$ of the circular motion. The second law in S' reads

$$m\,\vec{a}\,' = \sum_i \vec{F}_i \underbrace{- m\vec{A}}_{\vec{F}_{\text{fict}}}, \qquad \text{with } \sum_i \vec{F}_i = \vec{F}_c = \vec{F}_{\text{friction}}.$$

In S' the passenger is at rest, $\vec{a}\,' = 0$, therefore

$$\vec{F}_{\text{fict}} + \vec{F}_c = 0 \quad \Longrightarrow \quad \vec{F}_{\text{fict}} = -\vec{F}_c.$$

This fictitious force is the **centrifugal force**:

Centrifugal force

$$\vec{F}_{\text{cent}} = \vec{F}_{\text{fict}}, \qquad |\vec{F}_{\text{cent}}| = |\vec{F}_c| = m\,\frac{v^2}{r} = m\,\omega^2 r,$$

with the same magnitude as the centripetal force but opposite direction (away from the rotation axis). It appears only in a rotating reference frame.

Centripetal vs. centrifugal

- "The centrifugal force does not exist" means: it does not exist *in an inertial frame*, and nothing exerts it. In the rotating frame it is a perfectly useful (and necessary) term.
- *Centripetal* is not a new kind of force: it is the role played by real forces whose sum points to the centre. *Centrifugal* is a fictitious force. Never draw a centrifugal force in a free-body diagram in the inertial frame.
- If the string of a sling breaks, the stone flies off **along the tangent** (first law), not radially outwards as a "centrifugal push" would suggest.

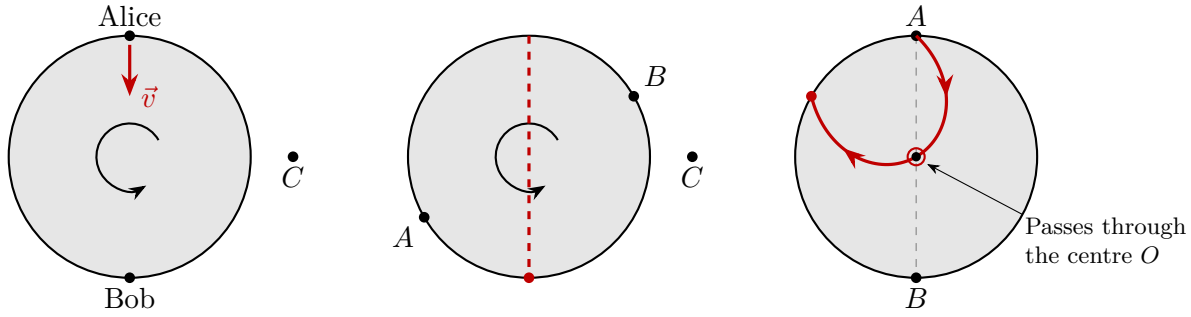
Worked example: how hard are you pushed against the door?

A car takes a curve of radius $r = 30\text{m}$ at $v = 15\text{m/s}$ (54km/h). Then $v^2/r = 225/30 = 7.5\text{m/s}^2 \approx 0.76\,g$. For a passenger of 70kg, the horizontal force that the seat and door must supply is about 525N. If static friction on the seat cannot provide it ($\mu_s < 0.76$), she slides until the door pushes her.

14.4 The Coriolis force

The centrifugal force acts on bodies *at rest* in the rotating frame. If a body *moves* in a rotating frame, a second fictitious force appears: the **Coriolis force**.

A circular platform rotates counterclockwise with angular velocity ω. Alice and Bob sit at opposite points of the edge; Charlie stands on the ground, in an inertial frame. Alice throws a ball with velocity $\vec{v}$ straight towards Bob.



(a) Alice throws; C = Charlie (ground)

(b) Charlie (inertial): straight line

(c) platform frame: curved trajectory

Charlie's point of view (inertial frame). After it leaves Alice's hand, no horizontal force acts on the ball: it moves with constant $\vec{v}$ on a straight line. Meanwhile the platform rotates, and when the ball reaches the edge, Bob is no longer there (panel b). If this were true also for Alice, she would expect Bob to receive the ball.

Platform reference frame. For Alice and Bob, who do not move on the platform, the trajectory is *not straight* (panel c): the ball bends sideways. They observe a change of $\vec{v}$ without a physical force. In the rotating frame this is attributed to a fictitious force acting on the moving ball: the **Coriolis force**. For a counterclockwise rotation the deflection is always **to the right** of the direction of motion (for a clockwise rotation, to the left).

Coriolis vs. centrifugal

The Coriolis force acts **only on bodies that move** in the rotating frame ($F_C = 0$ if $v' = 0$); the centrifugal force acts on all bodies, moving or not. Being always perpendicular to $\vec{v}\,'$, the Coriolis force changes the direction of the velocity, not the speed.

14.5 Is the Earth an inertial frame?

The Earth rotates about its axis and orbits the Sun, so it is not, strictly speaking, an inertial reference frame. But the corrections are small for most laboratory experiments:

Effect	acceleration	compared to g
rotation, centripetal at the equator, $\omega^2 R_E$	$\approx 0.034\text{m/s}^2$	0.3%
orbit around the Sun	$\approx 0.006\text{m/s}^2$	0.06%
Coriolis, $v = 10\text{m/s}$ in Palermo	$\approx 9 \times 10^{-4}\text{m/s}^2$	0.01%

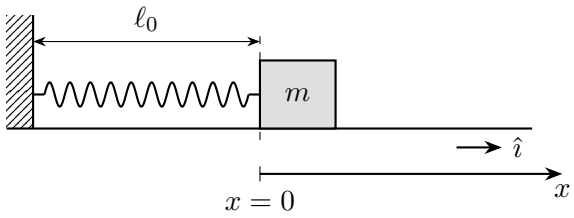
Coriolis effects become important on large scales and long times: the deflection of winds (hence the rotation of hurricanes, counterclockwise in the Northern Hemisphere), ocean currents, and the slow rotation of the plane of oscillation of the Foucault pendulum.

15 Motion of an object attached to a spring

At the very beginning of dynamics we used springs to show that forces are vectors, assuming that the elongation is proportional to the force. We now study the spring force properly. To deform a spring, stretching or compressing it, we must exert a force on it, and the spring reacts with a force of its own. We want to find its *force law*.

15.1 The physical system

Our system is a block of mass m on a horizontal table, attached to a wall by a spring.



If the object is at rest and nobody touches it, there is no deformation, and we say that the system is at **equilibrium**. The spring then has its **rest length** ℓ_0. We make the following idealisations:

- the table is **frictionless**;
- the spring has **no mass**;
- in the vertical direction the weight and the normal force cancel, so only horizontal forces matter.

We take the x axis along the spring, with unit vector $\hat{\imath}$ pointing away from the wall, and we put the origin $x = 0$ at the point at which the box is at equilibrium. With this choice, a displacement of the object Δx is exactly the deformation of the spring:

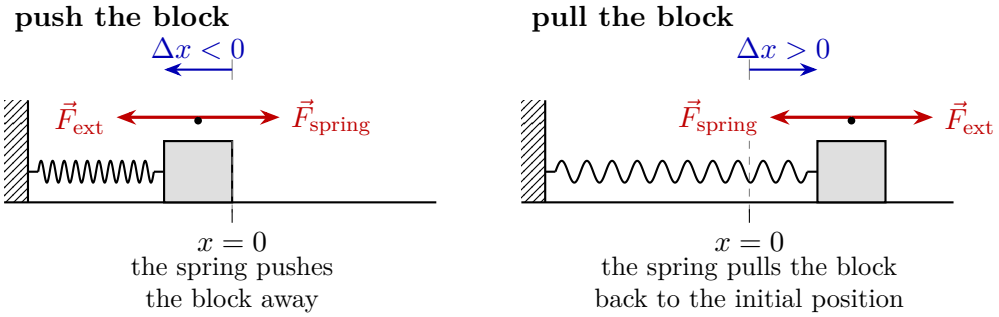
$$\Delta x = x = \ell - \ell_0, \qquad \begin{cases} \Delta x > 0 & \text{spring stretched,} \\ \Delta x < 0 & \text{spring compressed.} \end{cases}$$

Deformation, not length

x is the **deformation** (change of length), not the length of the spring. A spring of length $\ell = \ell_0$ exerts no force, however long ℓ_0 is.

15.2 Direction of the spring force

Now we push or pull the block with our hand (external force $\vec{F}_{\text{ext}}$) and hold it at rest in the deformed position.



Since the block is at rest, the net force on it vanishes:

$$\vec{F}_{\text{ext}} + \vec{F}_{\text{sp}} = 0 \quad \Longrightarrow \quad \vec{F}_{\text{sp}} = -\vec{F}_{\text{ext}}.$$

When we push (compress, $\Delta x < 0$), the spring pushes the block away from the wall; when we pull (stretch, $\Delta x > 0$), the spring pulls the block back. We conclude that the direction of the force exerted by the spring is always **towards the equilibrium point** $x = 0$, i.e. always **opposite to the displacement**:

$$\Delta x > 0 \text{ (stretched)} \;\Rightarrow\; F_{\text{spring}} < 0, \qquad \Delta x < 0 \text{ (compressed)} \;\Rightarrow\; F_{\text{spring}} > 0.$$

$\vec{F}_{\text{ext}}$ and $\vec{F}_{\text{sp}}$ are not a third-law pair

Both forces act **on the block**: they cancel only because the block is in equilibrium. The third-law partner of $\vec{F}_{\text{sp}}$ (spring on block) is the force of the *block on the spring*.

15.3 Magnitude of the spring force: an experiment

The equilibrium argument gives the direction, but not how large the force is. To study the magnitude $|\vec{F}_{\text{spring}}|$ we use the second law, $|\vec{F}_{\text{spring}}| = m|\vec{a}|$. We compress or stretch the spring by a known amount Δx, hold the block at rest, and release it. *At the instant of release* the

only horizontal force is the spring force, so measuring the acceleration $\vec{a}_0$ just after the release measures the force. Repeating at different Δx, one finds

$$m\,|\vec{a}_0| \;\propto\; |\Delta x|.$$

The constant of proportionality is the **spring constant** k:

$$|\vec{F}_{\text{sp}}| = k\,|\Delta x|,$$

with direction opposite to $\Delta\vec{x}$.

Measure at the release

As soon as the block moves, Δx changes, so the force and the acceleration change too: the motion is **not** uniformly accelerated. With a different mass $|\vec{a}_0|$ changes, but $m|\vec{a}_0|$ does not: k is a property of the spring alone.

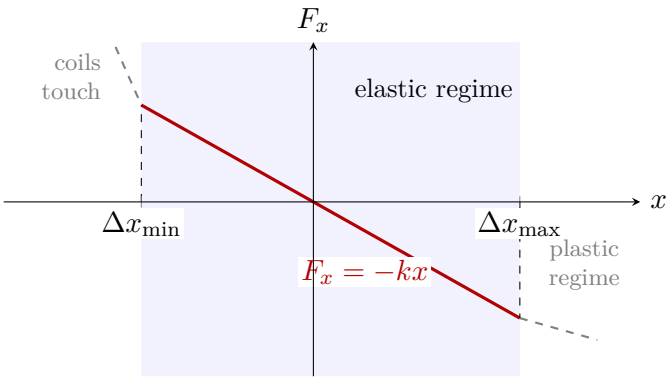
15.4 Hooke's law

Combining direction and magnitude:

Hooke's law

$$\vec{F}_{\text{sp}} = -k\,\Delta\vec{x}, \qquad \Delta\vec{x} = \Delta x\,\hat{\imath}\ (\Delta x \text{ positive or negative}), \qquad \Longrightarrow \qquad F_x = -k\,x.$$

The minus sign carries all the physics of the direction. A force always directed towards an equilibrium position is called a **restoring force**.



This law is valid under a few assumptions. It holds only for deformations in a range $\Delta x_{\text{min}} < \Delta x < \Delta x_{\text{max}}$, the **elastic regime**. If we stretch too much, the spring deforms permanently; if we compress too much, the coils touch. Hooke's law is an empirical, approximate law, not a fundamental law like Newton's laws. It is, however, extremely common: any force that vanishes at a stable equilibrium and varies smoothly around it looks, for small displacements, like $F \simeq -kx$. This is why the spring is the prototype of every oscillating system.

15.5 Units of the spring constant

The spring constant k is a property of the spring alone (material, thickness of the wire, number and size of the coils). From $|\vec{F}| = k|\Delta\vec{x}|$:

$$[F] = [k]\cdot[\Delta x] \quad \Longrightarrow \quad \frac{M\,L}{T^2} = [k]\;L \quad \Longrightarrow \quad [k] = \frac{M}{T^2},$$

measured in $\mathrm{N/m} = \mathrm{kg/s^2}$. k is the force we need to deform the spring by 1m: a large k means a stiff spring, a small k a soft one (a laboratory spring: 1–100N/m; a car suspension spring: 10^4–10^5N/m).

15.6 Examples

Worked example: holding and releasing the block

A spring with $k = 50\mathrm{N/m}$ is attached to a block of mass $m = 0.50\mathrm{kg}$ on a frictionless table.

(a) What $\vec{F}_{\mathbf{ext}}$ is needed to hold the block with $\Delta x = 4.0\mathrm{cm}$?

$$F_{\mathrm{sp},x} = -k\,\Delta x = -\left(50\tfrac{\mathrm{N}}{\mathrm{m}}\right)(0.040\mathrm{m}) = -2.0\mathrm{N},$$

$$\vec{F}_{\mathrm{ext}} + \vec{F}_{\mathrm{sp}} = 0 \quad \Longrightarrow \quad F_{\mathrm{ext},x} = -F_{\mathrm{sp},x} = 2.0\mathrm{N}:$$

$\vec{F}_{\mathrm{ext}}$ must pull the object with 2.0N.

(b) The block is released. What is $\vec{a}$ just after the release? $\vec{F}_{\mathrm{ext}}$ disappears instantaneously, so the second law gives $F_{\mathrm{sp},x} = m\,a_x$ and

$$a_x = \frac{F_{\mathrm{sp},x}}{m} = \frac{-k\,\Delta x}{m} = -\frac{(50\mathrm{N/m})(0.040\mathrm{m})}{0.50\mathrm{kg}} = -4.0\mathrm{m/s^2}:$$

the object accelerates in the negative direction, towards the wall.

(c) Same for $\Delta x = -2.0\mathrm{cm}$. We have a compression:

$$a_x = \frac{F_{\mathrm{sp},x}}{m} = \frac{-k\,\Delta x}{m} = -\frac{(50\mathrm{N/m})(-0.020\mathrm{m})}{0.50\mathrm{kg}} = +2.0\mathrm{m/s^2},$$

positive, and half the acceleration of (b): half the deformation gives half the force, and the sign flips.

Worked example: measuring k with a hanging mass

A spring hangs vertically from the ceiling. When a mass $m = 0.20\mathrm{kg}$ is attached and left at rest, the spring is stretched by $d = 5.0\mathrm{cm}$. At equilibrium the spring force balances the weight, $k\,d = mg$, so

$$k = \frac{mg}{d} = \frac{(0.20)(9.8)}{0.050}\mathrm{N/m} \approx 39\mathrm{N/m}.$$

This is the standard way to measure k in the lab, and how a dynamometer works.

15.7 Equation of motion

Equation of motion. The only horizontal force is the spring force (Hooke's law), $F_x = -kx$. Newton's second law along x gives

$$-kx = ma_x \qquad \Longrightarrow \qquad \boxed{a_x = -\frac{k}{m}\,x} \tag{1}$$

The acceleration of the block is

- **proportional** to its position x (i.e. to its displacement from equilibrium);
- directed **opposite** to the displacement (the minus sign).

Definition: simple harmonic motion

A body moves with **simple harmonic motion** (SHM) whenever its acceleration is proportional to its displacement from equilibrium and oppositely directed.

What happens, qualitatively. Pull the block to $x = A$ and release it from rest.

- At $x = A$: $v = 0$ and the acceleration is maximal and negative, $a_x = -kA/m$.
- The block speeds up toward the left. At $x = 0$ the force vanishes, so $a_x = 0$ and the speed is *maximum*.
- The block overshoots (by inertia!) and for $x < 0$ the force points to the right: the block slows down until it stops at $x = -A$, where $a_x = +kA/m$.
- It then comes back, passes through $x = 0$ with maximum speed and returns to $x = A$: one full **cycle**.

The points $x = \pm A$ are the **turning points**. Without friction, the motion repeats forever. Real systems always have some friction and the oscillations slowly die out (*damped* oscillations); here we deal with the ideal case.

Watch out!

The acceleration (1) changes with x, hence with time. This is **not** uniformly accelerated motion: the formulas $x = x_0 + v_0 t + \frac{1}{2}at^2$ and $v^2 = v_0^2 + 2a\Delta x$ **do not apply**. We need a new way to find $x(t)$.

15.8 The harmonic oscillator equation and its solution

Since $a_x = \mathrm{d}^2x/\mathrm{d}t^2$, equation (1) becomes

$$\boxed{\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\frac{k}{m}\,x(t) = -\omega^2\,x(t)}, \qquad \omega^2 \equiv \frac{k}{m}. \tag{2}$$

This is the **simple harmonic oscillator equation**. It is a *second-order differential equation*: the unknown is not a number but a *function* $x(t)$, and the equation tells us that its second derivative is equal to the function itself multiplied by a negative constant.

Guessing the solution. Which functions, differentiated twice, give back themselves with a minus sign? We know two: cos and sin. We try

$$\boxed{x(t) = A\cos(\omega t + \phi)} \tag{3}$$

with A, ω, ϕ constants. Differentiating:

$$v_x(t) = \frac{\mathrm{d}x}{\mathrm{d}t} = -\omega A\sin(\omega t + \phi), \tag{4}$$

$$a_x(t) = \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\omega^2 A\cos(\omega t + \phi) = -\omega^2\,x(t). \tag{5}$$

So (3) solves (2) **for any** A and ϕ, provided $\omega^2 = k/m$.

- A: **amplitude** (maximum displacement; we take $A > 0$);
- $\omega t + \phi$: **phase** of the motion (an angle, in radians);
- ϕ: **initial phase** (phase constant), the phase at $t = 0$;

- ω: **angular frequency**.

Why two free constants?

The equation of motion contains a *second* derivative. To determine the motion uniquely we must specify **two** initial conditions: the initial position $x(0)$ and the initial velocity $v_x(0)$. Correspondingly, the general solution contains two free constants, A and ϕ. The physics of the system (k and m) fixes ω; how we *start* the motion fixes A and ϕ.

15.9 Angular frequency, period and frequency

The cosine repeats itself when its argument increases by 2π. The time needed for this is the **period** T: $\omega(t+T)+\phi = \omega t + \phi + 2\pi$, so

$$\boxed{\omega = \sqrt{\frac{k}{m}}, \qquad T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}, \qquad f = \frac{1}{T} = \frac{\omega}{2\pi}} \tag{6}$$

The **frequency** f is the number of cycles per second, measured in hertz ($1\,\mathrm{Hz} = 1\,\mathrm{s}^{-1}$).

Dimensional check.

$$[\omega] = \frac{[k]^{1/2}}{[m]^{1/2}} = \sqrt{\frac{\mathrm{M\,T^{-2}}}{\mathrm{M}}} = \frac{1}{\mathrm{T}} \qquad \Rightarrow \qquad \omega \text{ is measured in rad/s.}$$

The radian is dimensionless; we write rad/s for ω and Hz for f to avoid confusing the two.

- A *stiffer* spring (larger k) gives faster oscillations (smaller T).
- A *heavier* block (larger m, more inertia) gives slower oscillations.
- The period does **not** depend on the amplitude A: large and small oscillations take the same time (*isochronism*). This is a special property of a restoring force *exactly proportional* to x.

Watch out

Here ω is **not** the angular velocity of something that rotates: the block moves on a straight line! It is just the constant that sets how fast the phase advances. We will see in Section 15.12 why it is natural to call it "angular". Also: ω and f differ by a factor 2π.

15.10 Velocity and acceleration

From (4) and (5), since $|\sin| \le 1$ and $|\cos| \le 1$,

$$\boxed{v_{\max} = \omega A = \sqrt{\frac{k}{m}}\,A, \qquad a_{\max} = \omega^2 A = \frac{k}{m}\,A} \tag{7}$$

Using $-\sin\alpha = \cos(\alpha + \pi/2)$ and $-\cos\alpha = \cos(\alpha+\pi)$ we can rewrite

$$x = A\cos(\omega t + \phi), \qquad v_x = \omega A\cos\left(\omega t + \phi + \tfrac{\pi}{2}\right), \qquad a_x = \omega^2 A\cos(\omega t + \phi + \pi).$$

The velocity is 90° **out of phase** with the position (it is zero when $|x|$ is maximum and maximal when $x = 0$); the acceleration is 180° **out of phase** with the position (it is always opposite to x).

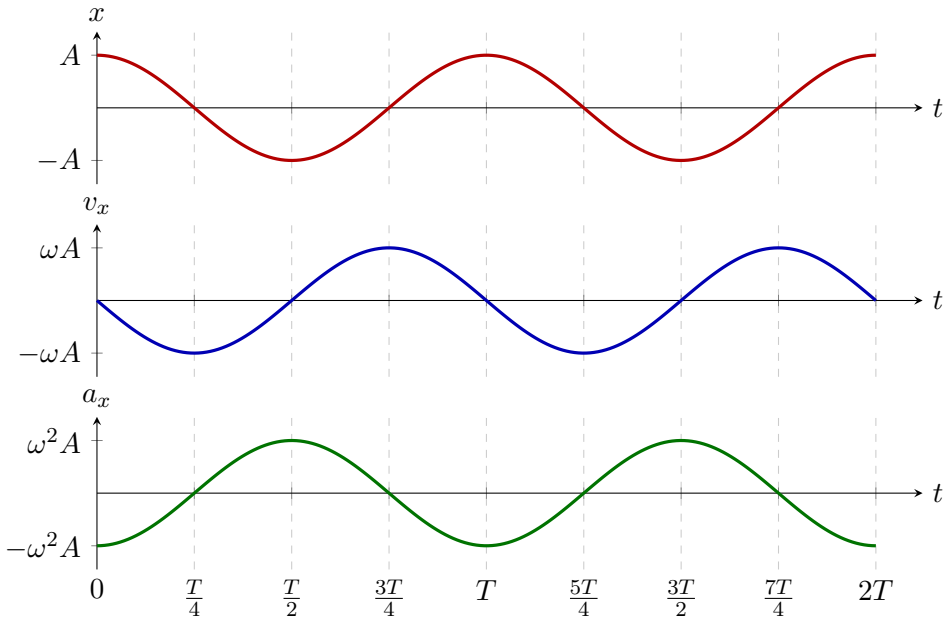


Figure 1: Position, velocity and acceleration for $\phi = 0$ (block released from rest at $x = A$). Where x is extreme, $v_x = 0$ and $|a_x|$ is maximum; where $x = 0$, $|v_x|$ is maximum and $a_x = 0$.

position	$x = A$	$x = 0$	$x = -A$
speed $\lvert v_x \rvert$	0	ωA (max)	0
acceleration a_x	$-\omega^2 A$	0	$+\omega^2 A$

15.11 Amplitude and phase from the initial conditions

Setting $t = 0$ in (3) and (4), with $x_0 \equiv x(0)$, $v_0 \equiv v_x(0)$:

$$x_0 = A\cos\phi, \qquad v_0 = -\omega A \sin\phi \quad \Longleftrightarrow \quad A\cos\phi = x_0, \qquad A\sin\phi = -\frac{v_0}{\omega}. \tag{8}$$

Amplitude. Square and add, using $\cos^2\phi + \sin^2\phi = 1$:

$$A^2 = x_0^2 + \frac{v_0^2}{\omega^2} \qquad \Longrightarrow \qquad \boxed{A = \sqrt{x_0^2 + \frac{v_0^2}{\omega^2}}} \tag{9}$$

Initial phase. Divide the second equation by the first:

$$\boxed{\tan\phi = -\frac{v_0}{\omega\, x_0}} \tag{10}$$

Watch out

- $\phi = \arctan(\dots)$ is only correct if $x_0 > 0$: the calculator returns an angle between $-\pi/2$ and $\pi/2$, but $\tan\phi$ does not change if we add π to ϕ. Always check with (8) that $\cos\phi$ has the **sign of** x_0 and $\sin\phi$ has the **sign of** $-v_0$; if not, add (or subtract) π.
- If $x_0 = 0$ the formula for $\tan\phi$ cannot be used: go back to (8) (see Example 2).

Worked example 1: released from rest

The most common case: the block is pulled to x_0 and released, $v_0 = 0$. Then $A = |x_0|$ and $\sin\phi = 0$.

- If $x_0 > 0$: $\cos\phi = 1$, $\phi = 0$, and $x(t) = A\cos\omega t$ (Fig. 1).
- If $x_0 < 0$: $\cos\phi = -1$, $\phi = \pi$, and $x(t) = A\cos(\omega t + \pi) = -A\cos\omega t$.

Worked example 2: a kick from the equilibrium position

At $t = 0$ the block passes through $x = 0$ moving to the right with speed v_i: $x_0 = 0$, $v_0 = v_i > 0$. From (8):

$$A\cos\phi = 0 \;\Rightarrow\; \phi = \pm\frac{\pi}{2}, \qquad A\sin\phi = -\frac{v_i}{\omega} < 0 \;\Rightarrow\; \phi = -\frac{\pi}{2}, \quad A = \frac{v_i}{\omega}.$$

Hence

$$x(t) = \frac{v_i}{\omega}\cos\left(\omega t - \frac{\pi}{2}\right) = \frac{v_i}{\omega}\sin\omega t.$$

The graphs are those of Fig. 1 shifted to the *right* by a quarter of a period: a phase of $-\pi/2$ is one fourth of a full cycle 2π, i.e. a delay of $T/4$.

15.12 Simple harmonic motion and uniform circular motion

A point P moves on a circle of radius A with **constant** angular speed ω. At time t the radius OP makes the angle $\theta(t) = \omega t + \phi$ with the x axis (ϕ is the angle at $t = 0$). Let Q be the projection of P on the x axis.

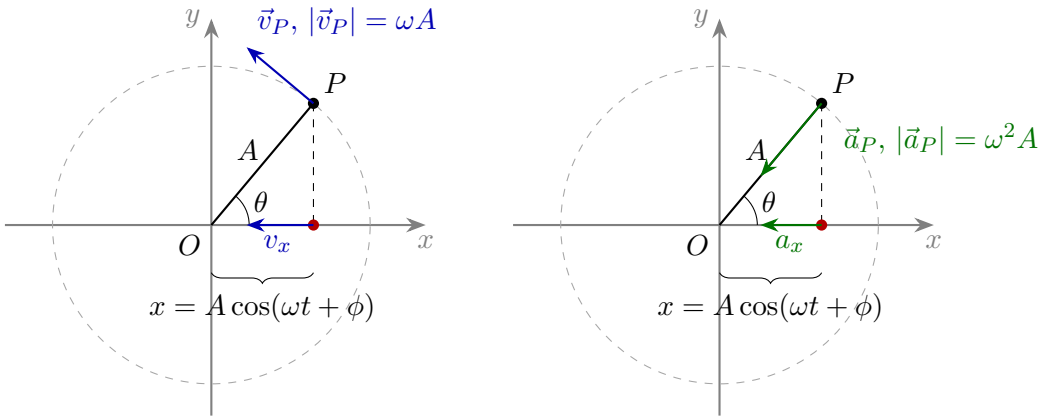


Figure 2: The projection Q of a point P in uniform circular motion, with $\theta = \omega t + \phi$. Left: the velocity of P is tangent to the circle; its x component is the velocity of Q. Right: the acceleration of P is centripetal; its x component is the acceleration of Q.

- **Position:** $x_Q = A\cos(\omega t + \phi)$.
- **Velocity:** $\vec{v}_P$ is tangent to the circle, with magnitude ωA; its x component is $v_x = -\omega A\sin(\omega t + \phi)$.
- **Acceleration:** $\vec{a}_P$ is centripetal, with magnitude $\omega^2 A$, pointing toward O; its x component is $a_x = -\omega^2 A\cos(\omega t + \phi) = -\omega^2 x_Q$.

These are exactly (3)–(5).

Key idea

The projection on a diameter of a uniform circular motion is a simple harmonic motion (and vice versa: every SHM can be seen as the "shadow" of a uniform circular motion, on a *reference circle*). The dictionary is:

angular speed ω of P	$\longleftrightarrow$	angular frequency ω of the SHM
radius of the circle	$\longleftrightarrow$	amplitude A
initial angle of OP	$\longleftrightarrow$	initial phase ϕ
time for one revolution	$\longleftrightarrow$	period T

This is why ω is called *angular* frequency.

Note also that the projection on the y axis, $y_Q = A\sin(\omega t+\phi) = A\cos(\omega t+\phi-\pi/2)$, is another SHM with the same ω, shifted by a quarter of a period.

16 The simple pendulum

16.1 The model

A **simple pendulum** is a body hanging from a string whose upper end is fixed at a pivot P. We idealize it as follows:

- the body is a **point mass** m (its size is negligible compared to L);
- the string is **massless** and **inextensible**, of length L;
- there is **no friction** (neither at the pivot nor with the air).

The body then moves along an arc of circle of radius L centred at P. Its position is described by a single variable: the angle θ between the string and the downward vertical ($\theta > 0$ on the right, $\theta < 0$ on the left).

16.2 Forces and equations of motion

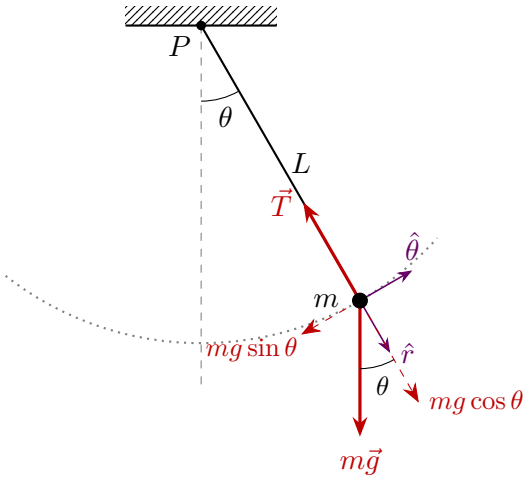


Figure 3: Forces on the pendulum bob: the tension $\vec{T}$ along the string and the weight $m\vec{g}$, decomposed along $\hat{r}$ and $\hat{\theta}$. The angle between $m\vec{g}$ and the string is θ.

Since the motion is on a circle, **polar coordinates** centred at the pivot are the natural choice:

$\hat{r}$ points along the string, from P toward the mass; $\hat{\theta}$ is perpendicular to it, in the direction of increasing θ.

The bob is forced to move on an arc of circle of radius L centred at the pivot: the string keeps it always at the same distance from P. Its position is therefore described by a single number, and it is convenient to look at the motion *along the arc*, i.e. along the direction $\hat{\theta}$, tangent to the circle and pointing toward increasing θ (Fig. 3).

Position, velocity and acceleration along the arc. Let s be the length of the arc from the lowest point to the bob, counted positive on the right ($\theta > 0$) and negative on the left. If θ is measured in **radians**, the definition of the radian gives

$$s(t) = L\,\theta(t).$$

Since L is constant, the velocity and the acceleration along the arc are

$$v_t = \frac{\mathrm{d}s}{\mathrm{d}t} = L\,\frac{\mathrm{d}\theta}{\mathrm{d}t}, \qquad a_t = \frac{\mathrm{d}v_t}{\mathrm{d}t} = \frac{\mathrm{d}^2 s}{\mathrm{d}t^2} = L\,\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2}.$$

a_t is the component of the acceleration along $\hat{\theta}$: it tells how fast the bob speeds up or slows down along its path.

Forces along the arc. Two forces act on the bob (Fig. 3):

- the **tension** $\vec{T}$, directed along the string toward P. It is perpendicular to the arc, so it has *no component* along $\hat{\theta}$: it cannot speed up or slow down the bob;
- the **weight** $m\vec{g}$. The angle between $m\vec{g}$ and the string is θ, so we split it into a part along the string, of magnitude $mg\cos\theta$, and a part along the arc, of magnitude $mg\sin\theta$. The latter points toward the lowest point, i.e. *opposite* to $\hat{\theta}$ when $\theta > 0$:

  $$F_t = -mg\sin\theta.$$

  The same expression holds for $\theta < 0$: then $\sin\theta < 0$ and $F_t > 0$.

The forces along the string (tension and $mg\cos\theta$) do not change the position of the bob along the arc; they only determine how hard the string pulls. To find the motion we only need the component along the arc.

Newton's second law along the arc. Projecting $\vec{T} + m\vec{g} = m\vec{a}$ on $\hat{\theta}$:

$$m\,a_t = F_t \qquad \Longrightarrow \qquad mL\,\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = -mg\sin\theta, \tag{11}$$

and dividing by mL,

$$\boxed{\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = -\frac{g}{L}\,\sin\theta} \tag{12}$$

The tangential force $-mg\sin\theta$ is a **restoring** force: for $\theta > 0$ it is negative, for $\theta < 0$ it is positive; it always pushes the bob back to $\theta = 0$. Notice that the mass has cancelled in (12): the motion of a pendulum **does not depend on its mass** (because the same m appears in the weight and in $m\vec{a}$).

16.3 Small oscillations

Equation (12) is *not* the harmonic oscillator equation: on the right there is $\sin\theta$, not θ. However, for small angles (**in radians**!)

$$\sin\theta \simeq \theta \qquad (|\theta| \ll 1).$$

θ (deg)	5°	10°	20°	30°	45°	60°
θ (rad)	0.0873	0.1745	0.3491	0.5236	0.7854	1.0472
$\sin\theta$	0.0872	0.1736	0.3420	0.5000	0.7071	0.8660
relative error	0.1%	0.5%	2%	5%	11%	21%

With this approximation,

$$\boxed{\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = -\frac{g}{L}\,\theta} \tag{13}$$

which is the harmonic oscillator equation (2) with $x \to \theta$ and $\omega^2 \to g/L$. We do not need to solve anything new:

$$\boxed{\theta(t) = \theta_{\max}\cos(\omega t + \phi), \qquad \omega = \sqrt{\frac{g}{L}}, \qquad T = 2\pi\sqrt{\frac{L}{g}}} \tag{14}$$

where $\theta_{\max}$ (the largest angle reached) and ϕ are fixed by the initial conditions exactly as in Section 15.11. For a pendulum released from rest at θ_0: $\theta_{\max} = \theta_0$ and $\phi = 0$.

A few observations:

- The period depends only on L and g: not on the mass, and (for small oscillations) not on the amplitude. A pendulum is therefore a good clock, and a simple way to *measure* g.
- Example: $L = 1.00\,\mathrm{m}$, $g = 9.81\,\mathrm{m/s^2}$ gives $T \simeq 2.01\,\mathrm{s}$.
- For larger amplitudes the period becomes longer: $T/T_{\mathrm{small}} \simeq 1.002$ at 10°, 1.017 at 30°, 1.073 at 60°. The motion is still periodic, but no longer a pure cosine.

16.4 Angular velocity of the pendulum

The angular velocity of the string (along z, perpendicular to the plane of motion) is

$$\omega_z(t) = \frac{\mathrm{d}\theta}{\mathrm{d}t} = -\sqrt{\frac{g}{L}}\,\theta_{\max}\,\sin\!\left(\sqrt{\frac{g}{L}}\,t + \phi\right),$$

and the speed of the bob is $v = L\,|\omega_z|$, maximum at the bottom: $v_{\max} = L\,\omega\,\theta_{\max} = \theta_{\max}\sqrt{gL}$.

Watch out

Two different "omegas" appear here:

- $\omega = \sqrt{g/L}$ is the **angular frequency**: a constant, related to the period.
- $\omega_z = \mathrm{d}\theta/\mathrm{d}t$ is the **angular velocity** of the string: it changes in time, vanishes at the turning points and is maximum ($\omega\,\theta_{\max}$) at the bottom.

They have the same units (rad/s) but are different physical quantities.